104 (2025) 105811

Contents lists available at ScienceDirect

Poultry Science

journal homepage: www.elsevier.com/locate/psj

Review

# Antimicrobial resistance in meat and meat products from Asia: An urgent one health challenge- a systematic review and meta-analysis

Md Jisan Ahmed [a,b,*], Md Imran Hossain [a,b], Md Ismile Hossain Bhuiyan [a,c], Md Arifur Rahman [a,d], Israt Jahan Kaderi [a,e], Ahmed Ishtiak Nirob [a], Md Afiqul Islam Rahi [a], Mustakim Mamun [a,c], Kazi Estieque Alam [a], Md Jayed Chowdhury [a], Prajwal Bhandari [a,f], Ritu Chalise [a,f], Md Jahangir Alam [c], Md Tazul Islam Chowdhury [a,g], Mahabbat Ali [a,c], Delower Hossain [a,h]

[a] *Association of Coding, Technology, and Genomics (ACTG), Sher-e-Bangla Agricultural University (SAU), Dhaka 1207, Bangladesh*
[b] *Department of Pathology, Faculty of Animal Science and Veterinary Medicine, Sher-e-Bangla Agricultural University (SAU), Dhaka 1207, Bangladesh*
[c] *Department of Animal Production and Management, Faculty of Animal Science and Veterinary Medicine, Sher-e-Bangla Agricultural University (SAU), Dhaka 1207, Bangladesh*
[d] *Department of Surgery and Theriogenology, Faculty of Animal Science and Veterinary Medicine, Sher-e-Bangla Agricultural University (SAU), Dhaka 1207, Bangladesh*
[e] *Department of Anatomy, Histology, and Physiology, Faculty of Animal Science and Veterinary Medicine, Sher-e-Bangla Agricultural University (SAU), Dhaka 1207, Bangladesh*
[f] *Department of Microbiology and Parasitology, Faculty of Animal Science and Veterinary Medicine, Sher-e-Bangla Agricultural University (SAU), Dhaka 1207, Bangladesh*
[g] *Department of Agricultural Chemistry, Faculty of Agriculture, Sher-e-Bangla Agricultural University (SAU), Dhaka 1207, Bangladesh*
[h] *Department of Medicine and Public Health, Faculty of Animal Science and Veterinary Medicine, Sher-e-Bangla Agricultural University (SAU), Dhaka 1207, Bangladesh*

A R T I C L E I N F O

*Keywords:*
Antimicrobial resistance
AMR gene
Meta-analysis
Meat
Asia

A B S T R A C T

*Background:* The increasing demand for animal protein poses a significant public health challenge because of the associated burden of foodborne diseases and antimicrobial resistance.
*Objective:* This study aimed to address the pooled prevalence of bacterial pathogens and their AMR patterns with associated genes found in meat and its products across Asia.
*Methods:* One hundred thirty-four relevant studies published between 2015 and 2025 were identified using major scientific electronic databases and selected on the basis of the inclusion criteria. Data were extracted in Microsoft Excel, and STATA (v19) and R programming (v4.4.2) were used for the analyses. A random effects model was applied for the pooled prevalence at the 95 % CI to account for substantial heterogeneity.
*Results:* The pooled prevalence of *Escherichia coli* (*E. coli*), *Salmonella* spp., *Staphylococcus aureus* (*S. aureus*), *Klebsiella* spp., *Campylobacter* spp., and *Enterococcus* spp. isolates from the meat and meat products were 49.6 %, 31.7 %, 38.1 %, 21.3 %, 30.7 %, and 46.9 %, respectively. The highest prevalence of isolated bacteria was from pigeons (39.3 %), followed by chickens. Similarly, *E. coli* was the most frequent pathogen and was highly resistant to ampicillin (79.7 %). *S. aureus* (84.7 %) and *Klebsiella* spp. (97.7 %) also exhibited strong resistance to ampicillin, while *Salmonella* spp. showed high resistance to oxytetracycline (71.3 %). Moreover, the most prevalent AMR genes across the studies were *tetA, blaTEM, blaCTX, sul1, blaSHV*, and *mcr1*.
*Conclusion:* The findings revealed increasing pathogen resistance to common antibiotics, highlighting the need for better antibiotic stewardship, monitoring, public awareness, and effective treatment strategies to address this growing threat.

* Corresponding author at: Association of Coding, Technology, and Genomics (ACTG), Sher-e-Bangla Agricultural University (SAU), Dhaka 1207, Bangladesh.
*E-mail address:* jisan.path1506605@sau.edu.bd (M.J. Ahmed).

https://doi.org/10.1016/j.psj.2025.105811
Received 6 August 2025; Accepted 9 September 2025
Available online 10 September 2025

## Introduction

Global antimicrobial resistance (AMR) has become a significant threat to the effective prevention and management of current bacterial epidemics (Nwobodo et al., 2022; Ahmed et al., 2024). AMR has been listed as one of the 10 worldwide health risks to humans (Kumar, 2021). It was identified as being responsible for approximately 1.27 million deaths internationally in 2019, and that pattern will endure if no corrective measures are initiated (Kalpana et al., 2024). Moreover, the annual impact of AMR-related deaths might eclipse 10 million by 2050, exceeding the potential danger present in cancers (Murray et al., 2022). Furthermore, resistant microbes may access the food chain via ingestion, increasing the probability of AMR in foodborne infections (Thapa et al., 2020).

The Asia–Pacific region (APAC), which is the residence of two-thirds of the global population and ten of the most impoverished nations, is extremely susceptible to AMR-related concerns (Yam et al., 2019). AMR is predicted to increase by 70 % across Asia, posing a danger both nationally and globally (Kang and Song, 2013). AMR hot spots can be found on global maps of Asia, including the northeast region of India, China's northeast region, northern Pakistan, Iran, eastern Turkey, and the area around the Red River in Vietnam (Van Boeckel et al., 2019). Meat is classified as the primary source of foodborne pathogen transmission to humans (Ali and Alsayeqh, 2022). Pork is the most popular red meat in Asian food, particularly in South Korea, Vietnam, and China. From 2022 to 2029, pork consumption is projected to increase by 2 % in South Korea and 22 % in China and Vietnam. Moreover, poultry meat is consumed more per capita than both beef and sheep meat across the region (AHBD, 2025). These meats can carry AMR genes, posing risks to human health. Approximately 40 % of AMR genes are transmitted between humans and pork, whereas 24.7 % are associated with poultry (Cao et al., 2022).

The microbial content of meat and its products is affected by various factors, including animal management before slaughter, temperature control, preservation methods, packaging, and consumer practices (Odeyemi et al., 2020). Moreover, inadequate slaughterhouse infrastructure and poor hygiene during meat handling and transportation can result in microbial contamination, which spreads throughout the food chain, affecting both formal and informal markets (Rortana et al., 2021). The trends in prevalence of AMR highlight the necessity of a One Health approach, which includes surveillance, research, and awareness efforts to address the challenges of resistant pathogens (Nhung et al., 2024).

Meat-borne diseases worldwide are frequently caused by pathogens such as *Enterobacteriaceae, Salmonella* spp., *Listeria* spp., *Campylobacter* spp., *Brucella* spp., and toxins from *Staphylococcus* spp., *Clostridium* spp., and *Bacillus* spp. (Ali and Alsayeqh, 2022). *Escherichia coli* (*E. coli*), *Salmonella*, and *S. aureus* are frequently found as pathogenic bacteria in chicken, beef, and other meat products. (Abd El Tawab et al., 2015). *E. coli* strains can lead to several illnesses in humans, such as urinary tract infections, septicemia, meningitis, endocarditis, enteritis, hemolytic-uremic syndrome, and epidemic diarrhea (Islam et al., 2021). Salmonellosis in humans is frequently caused by non-typhoidal *Salmonella enterica serovars*, which are a major source of foodborne infections worldwide (Kirk et al., 2015). *S. aureus* is the most pathogenic staphylococcal and is responsible for a variety of diseases in humans and animals, including skin infections, pneumonia, septicemia, toxic shock syndrome, food poisoning, and endocarditis (Silva et al., 2023). Hyper-virulent *Listeria monocytogenes* clonal complexes in pig tonsils can cause human listeriosis (Oswaldi et al., 2022). Understanding the AMR profile of these pathogens is vital, as they are major contributors to hospital-acquired, community-based, and zoonotic infections globally.

The use of antibiotics in food production has made food animals key reservoirs of antibiotic-resistant bacteria, which worsens the AMR crisis (Founou et al., 2016). According to a study, 13 different classes of antibiotics were found to be ineffective against *E. coli* isolated from poultry meat (Parvin et al., 2020). In beef samples, *Salmonella* isolates showed a strong resistance to antibiotics, including ampicillin, ciprofloxacin, neomycin, tetracycline, amoxicillin, and oxytetracycline (Hussain et al., 2020). *Listeria* spp., *Campylobacter* spp., and *Clostridium* spp. are also known to exhibit antibiotic resistance in various meat and meat products (Andrzejewska et al., 2019). Furthermore, methicillin-resistant *S. aureus* (MRSA) is a rising zoonotic threat to both public health and veterinary care (Algammal et al., 2020). The livestock and poultry carry similar mobile resistance genes found in various animal-derived microbial strains, which contribute to drug resistance in humans (Li et al., 2019). These genes can spread horizontally and vertically among different bacterial species, potentially reaching the human food supply through the consumption of meat contaminated with resistant microorganisms (Hazards et al., 2021; Conceição et al., 2023). The rise of multidrug-resistant (MDR) strains in recent decades has become a major concern, with food animals and their products serving as key carriers of resistance genes (Khan et al., 2020).

The contamination of meat and meat-based products by foodborne pathogens is a critical public health issue, as these microorganisms pose health risks and result in significant socioeconomic effects (Zelalem et al., 2019). This alarming trend highlights the urgent need for comprehensive research on AMR, as literature often falls short, focusing on either a specific bacterial species or antibiotic rather than considering broader clinical and public health implications. To date, no meta-analysis has specifically addressed resistance genes in meat and meat products from retail markets, processing plants, and slaughterhouses across the entire Asian continent. However, a general study exists for Southeast Asia and South Asia, but it includes only limited data on meat (Nhung et al., 2016; Harun et al., 2024). Therefore, this systematic review and meta-analysis were conducted to investigate bacterial pathogen prevalence, resistance gene presence, and AMR patterns in meat- and meat-based products sourced from Asian retail and abattoir settings.

## Materials and methods

### *Guidelines and protocol*

The study adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines for conducting a systematic review and meta-analysis. To ensure the inclusion of all relevant information and uphold methodological standards, the PRISMA 2009 checklist was followed (S1 Table 1).

### *Systematic literature search strategy*

A comprehensive literature review was performed for this systematic review and meta-analysis to identify peer-reviewed studies reporting data on antibiotic-resistant organisms in meat from various species across Asian countries. Seven authors (MIH, MIHB, MAR, IJK, MAIR, MM, and AIN) rigorously investigated PubMed, Google Scholar, OpenAlex, and Science Direct for relevant peer-reviewed articles reporting primary research conducted from January 1, 2015, to May 11, 2025. A Boolean search strategy was employed, utilizing terms associated with AMR, meat, poultry, AMR genes, livestock, and Asian countries (S1 Table 2). These terms were informed by relevant research, systematic reviews, reports, and expert insights from veterinary medicine and antibiotic resistance specialists (Nellums et al., 2018; Nisansala et al., 2025). The database search was conducted with filters based on the inclusion and exclusion criteria to ensure relevance.

### *Inclusion and exclusion criteria*

For this systematic review and meta-analysis, specific inclusion and exclusion criteria were established to ensure the selection of relevant, high-quality studies on antibiotic-resistant organisms and AMR genes in meat from various species in Asian countries. The included studies were peer-reviewed, English-language articles reporting primary research conducted between January 1, 2015, and May 11, 2025; retrieved from PubMed, Google Scholar, OpenAlex, and ScienceDirect; and focused on antibiotic resistance or AMR genes in meat samples from poultry or livestock in Asian countries. The excluded studies included non-primary studies (e.g., reviews, editorials), non-English publications, studies conducted outside Asia, studies published outside the specified date range, or those not addressing antibiotic resistance or AMR genes in meat, ensuring the rigor and relevance of the findings.

*Data collection and extraction*

Seven authors (MIH, MIHB, MAR, IJK, MAIR, MM, and AIN) independently worked on each stage of the review process, including screening titles and abstracts, reviewing full texts, extracting data, and performing quality assessment. Discrepancies in study selection or data extraction were resolved through discussion between the authors, and when necessary, a third senior author (DH) was consulted to reach a final decision. The screening process began with the evaluation of titles and abstracts, followed by full-text assessment. Articles were included if they reported the sample type (meat), source (species), or country of origin within Asia. Given the common use of specific antibiotics in Asia, data extraction was directed toward resistance information for the following antibiotics: amoxicillin, ampicillin, azithromycin, cefotaxime, ceftriaxone, chloramphenicol, erythromycin, streptomycin, tetracycline, TMP-SMZ, gentamicin, amikacin, levofloxacin, colistin, nalidixic acid, doxycycline, lincomycin, neomycin, penicillin, kanamycin, cefuroxime, vancomycin, cefoxitin, oxacillin, and oxytetracycline. These antibiotics are classified into antibiotic classes, including aminoglycoside (amikacin, gentamicin, kanamycin, neomycin, streptomycin), beta-lactam (amoxicillin, ampicillin, oxacillin, penicillin), macrolide (azithromycin, erythromycin, tylosin), cephalosporin (cefotaxime, cefoxitin, Ceftriaxone, cefuroxime), phenicol (chloramphenicol), fluoroquinolone (ciprofloxacin, enrofloxacin, levofloxacin), polymyxin (colistin), tetracycline (doxycycline, tetracycline, oxytetracycline), lincosamide (lincomycin), quinolone (nalidixic acid), sulfonamide (trimethoprim-sulfamethoxazole), and glycopeptide (vancomycin) (S2 Table 1, 2, and 3).

*Bias assessment and quality evaluation*

In this systematic review, the methodological quality and risk of bias of the included studies were assessed via the Joanna Briggs Institute critical appraisal tools, which are tailored to specific study designs and focus on evaluating internal validity by identifying potential sources of bias in study design, conduct, and analysis (Grammatopoulos et al., 2023). The assessment encompassed eight key domains: appropriateness of the research design, clarity of the study setting, adequacy of population demographics, quality of sample data, procedures for sample transfer and storage, use of microbiological media, reporting of isolates, and transparency of data sources. Each criterion was evaluated via a binary scoring system: a score of '1′ was assigned for 'yes' responses indicating adherence to the criterion, and '0′ was assigned for 'no' responses indicating nonadherence. The cumulative score for each study ranged from 0 to 8 (S2 Table 4). On the basis of the total score, studies were categorized into three levels of bias risk: low risk of bias (scores above 70 %), moderate risk of bias (scores between 50 % and 69 %), and high risk of bias (scores below 50 %). To ensure the robustness of the findings, sensitivity analyses were conducted to examine the impact of excluding studies with a high and moderate risk of bias on the overall results. This structured approach to quality assessment enhances the reliability of the review by systematically identifying and accounting for potential biases in the included studies.

*Statistical analysis*

The retrieved data were then entered into an Excel spreadsheet. As part of this comprehensive meta-analysis, the authors meticulously reviewed and assessed each article that met the inclusion criteria to ensure the integrity and reliability of the synthesized findings. By using a random effects model meta-analysis, the pooled prevalence of bacteria and AMR patterns at the 95 % CI were determined (DerSimonian and Laird, 1986; Abebaw, 2024). Heterogeneity is classified as low, moderate, or high at $I^2$ values of 25 %, 50 %, and 75 %, respectively, with 0 % indicating its absence (Higgins and Thompson, 2002). A random-effects meta-analysis was chosen for summary statistics because of the substantial heterogeneity across studies. A subgroup analysis was conducted by country, source, species, and sample. Each categorical variable was dummy-coded (with k levels, we created $k-1$ binary indicators) to allow proper inclusion in meta-regression models (Alkharusi, 2012). Moreover, univariate meta-regression was conducted to explore potential sources of heterogeneity. Multivariate meta-regression was not pursued due to the risk of overfitting, given the limited number of studies within specific subgroups and the overall number of predictors relative to sample size (Harrer et al., 2021). The genotypic prevalence and antibiotic resistance patterns were visualized via a heatmap. The pooled prevalence of isolated bacteria is displayed on a map. Funnel plots and sensitivity analyses were employed for visual inspection of distribution, quality, and bias across articles. Small-study effects were assessed using Egger's regression test and Begg's rank correlation test. Egger's and Begg's tests provided statistical evaluation of potential small-study effects. The significance threshold was set at $p < 0.05$. Meta-analysis of the data, including meta-regression, subgroup analysis, forest plots, funnel plots, and sensitivity analysis, was conducted via Stata version 18.0 (College Station, 163 TX, USA). R programming (version 4.4.2) was used to create the heatmap and map, using different packages such as "*ggplot*" and "*spdep*" (Wickham, 2011; Bivand et al., 2017). This systematic review and meta-analysis were based on the PRISMA statement (Moher et al., 2009).

## Results

*Search results and eligible studies*

A PRISMA flow diagram outlining the study selection process is shown in Fig. 1. A total of 983 records were retrieved through database searches. After 129 duplicates were removed, 854 unique records remained for screening. On the basis of the title and abstract evaluation, 548 records were excluded. A total of 306 full-text articles were subsequently assessed for eligibility, of which 115 were excluded for reasons such as nonavailability in English, inaccessible full texts, or insufficiently detailed information. Ultimately, 134 studies met the inclusion criteria and were incorporated into the final qualitative synthesis.

*Descriptive characteristics of the included studies*

The study includes 134 published research articles sourced that met the eligibility criteria for inclusion in this systematic review and meta-analysis from 28 Asian countries; Bangladesh ($n = 23$) (Islam et al., 2018; Rahman et al., 2018; Neogi et al., 2020; Parvin et al., 2020; Rahman et al., 2020; Ali et al., 2021; Parvin et al., 2021; Rabby et al., 2021; Ain et al., 2022; Hossain et al., 2022; Islam et al., 2022; Julqarnain et al., 2022; Rahman and Ahmed, 2022; Samad et al., 2022; Alam et al., 2023; Ali et al., 2023; Samad et al., 2023; Al Naser et al., 2024; Hossain et al., 2024; Liza et al., 2024; Rafiq et al., 2024; Tanzin et al., 2024; Karim et al., 2025), China ($n = 13$) (Yu et al., 2015; Zhang et al., 2016; Wu et al., 2018, 2018; Yang et al., 2019; Zhou et al., 2019; Chen et al., 2020; Yang et al., 2020; Li et al., 2021; Wu et al., 2021; Hasib et al., 2024; Liu et al., 2024; Ren et al., 2024), Pakistan ($n = 11$) (Asif et al., 2017; Nisar et al., 2017; Samad et al., 2018; Azam et al., 2019; Khan

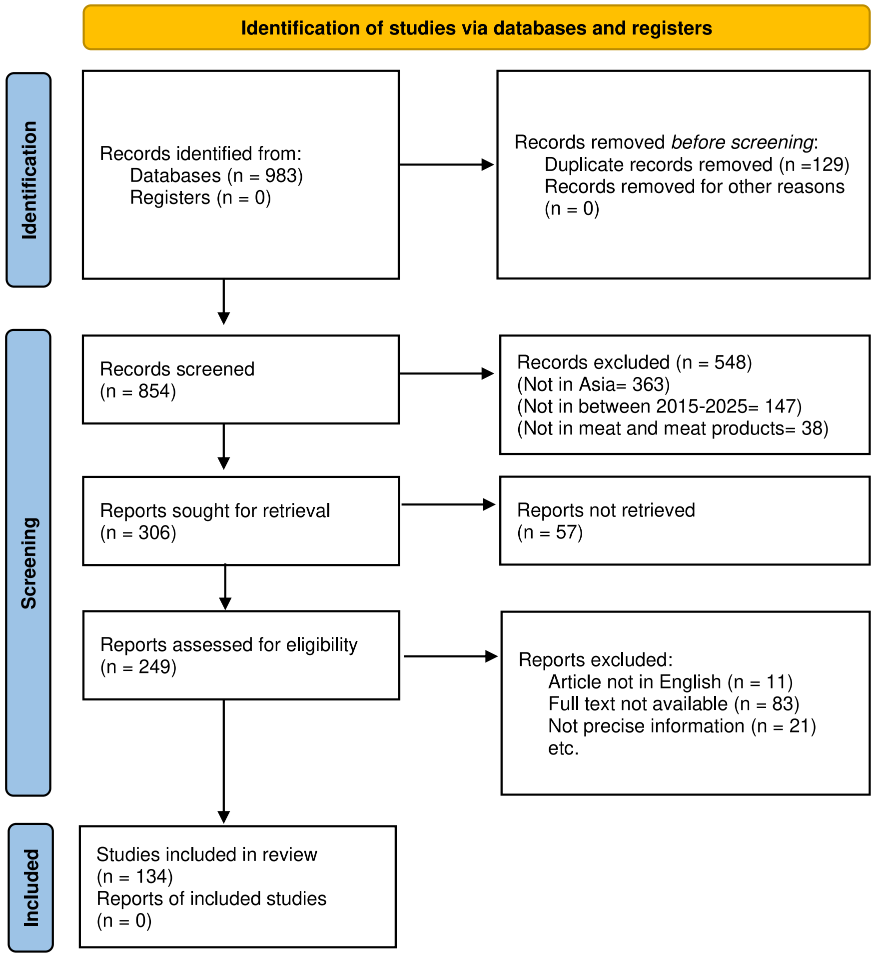


**Fig. 1.** PRISMA flow diagram for the selection of studies.

et al., 2019; Altaf Hussain et al., 2020; Sadiq et al., 2020; Liaqat et al., 2022; Amjad et al., 2023; Fatima et al., 2023; Shabbir et al., 2024), Nepal ($n = 8$) (Shrestha et al., 2017; Devkota et al., 2019; Joshi et al., 2019; Mahato, 2019; Saud et al., 2019; Bista et al., 2020; Nelson et al., 2020; Fowler et al., 2021), Malaysia ($n = 7$) (Thung et al., 2016; Premarathne et al., 2017, 2018; Tan et al., 2019; Zakaria et al., 2021; Aklilu et al., 2022; Karim et al., 2023), Turkey ($n = 7$) (Pehlivanlar Önen et al., 2015; Siriken et al., 2015; Baran et al., 2020; Adiguzel et al., 2021; Şanlıbaba, 2022; Kürekci et al., 2023; Aydin et al., 2024), India ($n = 6$) (Makwana et al., 2015; Naik et al., 2015; Kaushik et al., 2018; Savariraj et al., 2019; Khan et al., 2022; Priya et al., 2023), South Korea ($n = 6$) (Wei et al., 2016; Seo et al., 2018; Kim et al., 2019; Seo and Lee, 2019, 2020; Kang et al., 2024), Iraq ($n = 4$) (Sheet et al., 2021; Kanaan et al., 2022; Taib and Abdulrahman, 2022; Radi and Al-Marzoqi, 2023), Thailand ($n = 4$) (Kanokudom et al., 2021; Noenchat et al., 2022; Kansaen et al., 2023, 2023), Cambodia ($n = 4$) (Trongjit et al., 2017; Nadimpalli et al., 2019; Rortana et al., 2021; Huoy et al., 2024), Indonesia ($n = 3$) (Soepranianondo and Wardhana, 2019; Wardhana et al., 2021; Takaichi et al., 2022), Iran ($n = 3$) (Bokharaei et al., 2020; Gholami-Ahangaran et al., 2021; Mir et al., 2022), Japan ($n = 3$) (Furukawa et al., 2017; Mori et al., 2018; Asakura et al., 2022), Laos ($n = 3$) (Moser et al., 2021; Xaymountry et al., 2022; Zhou et al., 2022), Philippines ($n = 3$) (Santos et al., 2020; Belotindos et al., 2021; Nagpala et al., 2025), Saudi Arabia ($n = 3$) (El-Ghareeb et al., 2020; Al-Hindi et al., 2023; Alhadlaq et al., 2023), Sri Lanka ($n = 3$) (Kottawatta et al., 2017; Ranasinghe et al., 2022; Weerasooriya et al., 2024), United Arab Emirates (UAE) ($n = 3$) (Habib et al., 2022, 2023, 2024), Afghanistan ($n = 2$) (Shakhes et al., 2023; Ahmadi et al., 2025), Jordan ($n = 2$) (Tarazi et al., 2021; Al-Qadiri et al., 2025), Mongolia ($n = 2$) (Dorjgochoo et al., 2023, 2025), Myanmar ($n = 2$) (Moe et al., 2017; Lay et al., 2021), Qatar ($n = 2$) (Abu-Madi et al., 2016; Eltai et al., 2020), Russia ($n = 2$) (Zaiko et al., 2021; Rakitin et al., 2022), Singapore ($n = 2$) (Zwe et al., 2018; Guo et al., 2020), Vietnam ($n = 2$) (Nguyen et al., 2016; Le et al., 2025), and Kazakhstan ($n = 1$) (Mendybayeva et al., 2023) (Fig. 2).

*Pooled prevalence of pathogens*

The pooled prevalence rates of *E. coli, Salmonella* spp., *S. aureus, Klebsiella* spp., *Campylobacter* spp., and *Enterococcus* spp. isolates from the meat samples of different species were 49.6 % (95 % CI: 41.4–57.8 %, $p < 0.001$), 31.7 % (95 % CI: 25.3–38.5 %, $p < 0.001$), 38.1 % (95 % CI: 29.6–47 %, $p < 0.001$), 21.3 % (95 % CI: 3.6–47.8 %, $p < 0.001$), 30.7 % (95 % CI: 20.1–42.5 %, $p < 0.001$), and 46.9 % (95 % CI: 19.3–75.7 %, $p < 0.001$), respectively (Figs. 3–8).

Subgroup meta-analysis was conducted regardless of the types of isolated species on the basis of the countries in Asia, species, sources, and types of samples. Among the 28 countries, Myanmar presented the highest prevalence of bacteria isolated from meat, which was 77.8 % (95 % CI: 26.1–100 %, $p < 0.0001$), followed by the Philippines (62.1 %, 95 % CI: 49.5–73.8 %, $p < 0.001$), Sri Lanka (58.8 %, 95 % CI: 41.5–75.1 %, $p < 0.0001$), Cambodia (57.2 %, 95 % CI: 43.6–70.2 %, $p < 0.0001$), and Qatar (53.9 %, 95 % CI: 10.3–94.1 %, $p < 0.0001$). In terms of the sources of the meat samples, processing plants (40.3 %, 95 % CI: 20.9–61.5 %, $p < 0.0001$) presented the highest prevalence compared with farms (37.5 %, 95 % CI: 12.9–66.1 %, $p < 0.0001$), slaughterhouses (37.3 %, 95 % CI: 27.2–47.9 %, $p < 0.0001$), and markets (33.1 %, 95 %

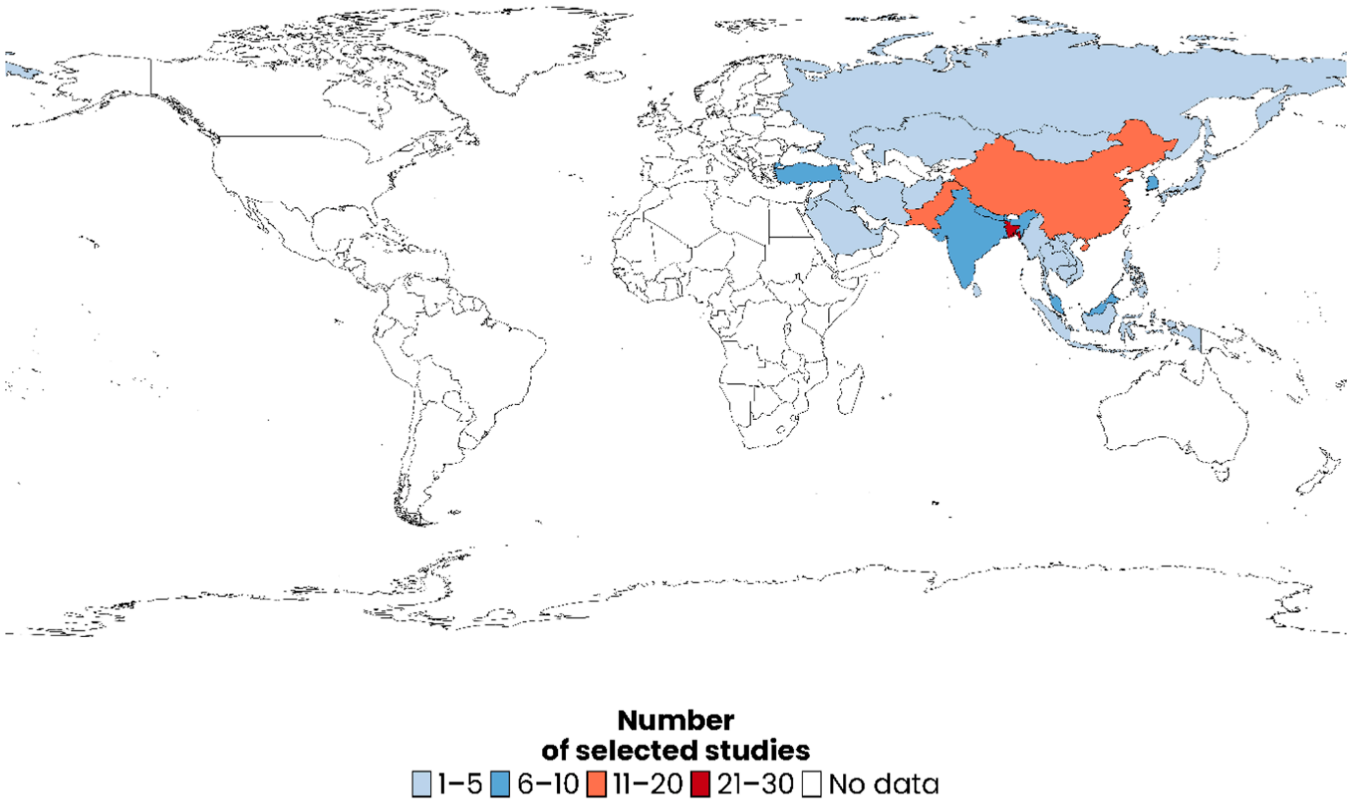


**Fig. 2.** A world map illustrating the geographical distribution of the selected studies for this meta-analysis ($n = 134$) was generated via the *ggplot2* package in R software (v 4.4.2) (Wickham, 2011).

CI: 29.4–36.9 %, $p < 0.0001$).

The meat sample sources included various animals and birds, such as cattle, buffalo, chicken, duck, goat, sheep, turkey, pig, pigeon, camel, and combinations of two or three species (mix). The bacterial species with the highest prevalence was pigeon (39.3 %, 95 % CI: 21.8–58.2 %), followed by chicken (37.5 %, 95 % CI: 32.9–42.4 %, $p < 0.0001$) and pig (35.2 %, 95 % CI: 24.4–46.7 %, $p < 0.0001$). Furthermore, the meat samples were classified into meat and meat products. Among them, meat (33.6 %, 95 % CI: 30–37.2 %, $p < 0.0001$) had a greater percentage of isolated bacterial species than did meat products (29.5 %, 95 % CI: 17.9–42.5 %, $p < 0.0001$) (Table 1).

The funnel plots (Figs. 9–11) revealed visual asymmetry, likely attributed to a small-study effect. Egger's regression test ($p > 0.05$) and Begg's rank correlation test ($p > 0.05$) were performed to assess potential small-study effects among the included studies. Both tests indicated no statistically significant evidence of publication bias (**S3** Fig. 1–12). Sensitivity analyses were performed across all six bacterial genera, including *E. coli, Salmonella* spp., *S. aureus, Klebsiella* spp., *Campylobacter* spp., and *Enterococcus* spp., to evaluate robustness, ensuring the stability and reliability of the meta-analysis. The pooled prevalence estimates remained consistent, ranging from 48 to 51 % for *E. coli*, 30–32 % for *Salmonella* spp., 36–40 % for *S. aureus*, 27–33 % for *Campylobacter* spp., 13–27 % for *Klebsiella* spp., and 32–57 % for *Enterococcus* spp., indicating stable and reliable outcomes (Figs. 12–14).

Owing to the marked heterogeneity among studies, univariate meta-regression was conducted to explore possible contributing factors. Among the covariates (study year, country, species, sample, source, sample size, and positive cases) examined for this meta-analysis, sample size ($\beta = 0.0012$, 95 % CI: 0.0005 to 0.0018, $p = 0.0005$) and sample size ($\beta = -0.0002$, 95 % CI: $-0.0003$ to 0.0001, $p = 0.004$) for *E. coli* were statistically significant (*$p < 0.05$*). Additionally, positive cases were statistically significant ($\beta = 0.0009$, 95 % CI: 0.0004 to 0.0013, $p = 0.0003$**)** for *Salmonella* spp. and ($\beta = 0.0044$, 95 % CI: 0.0018 to 0.0070, $p = 0.001$) for *Campylobacter* spp. Moreover, study year ($\beta = -0.0436$, 95 % CI: $-0.0824$ to $-0.0047$, $p = 0.028$) and sample ($\beta = -0.1525$, 95 % CI: $-0.2898$ to $-0.0151$, $p = 0.029$) for *S. aureus* were statistically significant *($p < 0.05$)*. Furthermore, the samples ($\beta = 1.3522$, 95 % CI: 0.3147 to 2.3897, $p = 0.011$) and positive cases ($\beta = 0.0392$, 95 % CI: 0.0253 to 0.0531, $p = 0.000$) were found to be significant for *Klebsiella* spp., and no covariates were found to be significant for *Enterococcus* spp. (Table 2).

The spatial distribution of the pooled prevalence for the six bacterial species, including *E. coli, Salmonella* spp., *S. aureus, Klebsiella* spp., *Campylobacter* spp., and *Enterococcus* spp., is illustrated (Fig. 15). Among Asian countries, higher prevalence rates of *E. coli*, ranging from 60 to 90 %, were observed predominantly in Southeast Asia and South Asia, including Myanmar, Cambodia, Thailand, the Philippines, Sri Lanka, Pakistan, and Afghanistan, as well as in the Middle Eastern country of Qatar. Similarly, *Salmonella* spp. were found to be highly prevalent in Myanmar, Cambodia, and Sri Lanka. The pooled prevalence was observed in South Asian countries, with *S. aureus* detected at 57 % in India, 45 % in Pakistan, 43 % in Bangladesh, and 39 % in Nepal, whereas a higher rate of 66 % was reported in Iraq. *Klebsiella* spp. were detected primarily in Bangladesh, followed by Nepal and Malaysia. *Enterococcus* spp. are most commonly reported in the UAE, with subsequent detection in Thailand and Bangladesh. For *Campylobacter* spp., the highest pooled prevalence was observed in South Korea, followed by Sri Lanka, Japan, and Bangladesh.

### *Meta-analysis of AMR carriage*

Among the six bacterial species analyzed, *E. coli* was the most frequently isolated organism, displaying variable resistance rates across antibiotic classes. The highest resistance rates were detected for ampicillin (79.7 %, 95 % CI: 66.6–90.2 %), followed by penicillin (75.6 %, 95 % CI: 13.8–100 %), oxytetracycline (72.8 %, 95 % CI: 32.9–98.8 %),

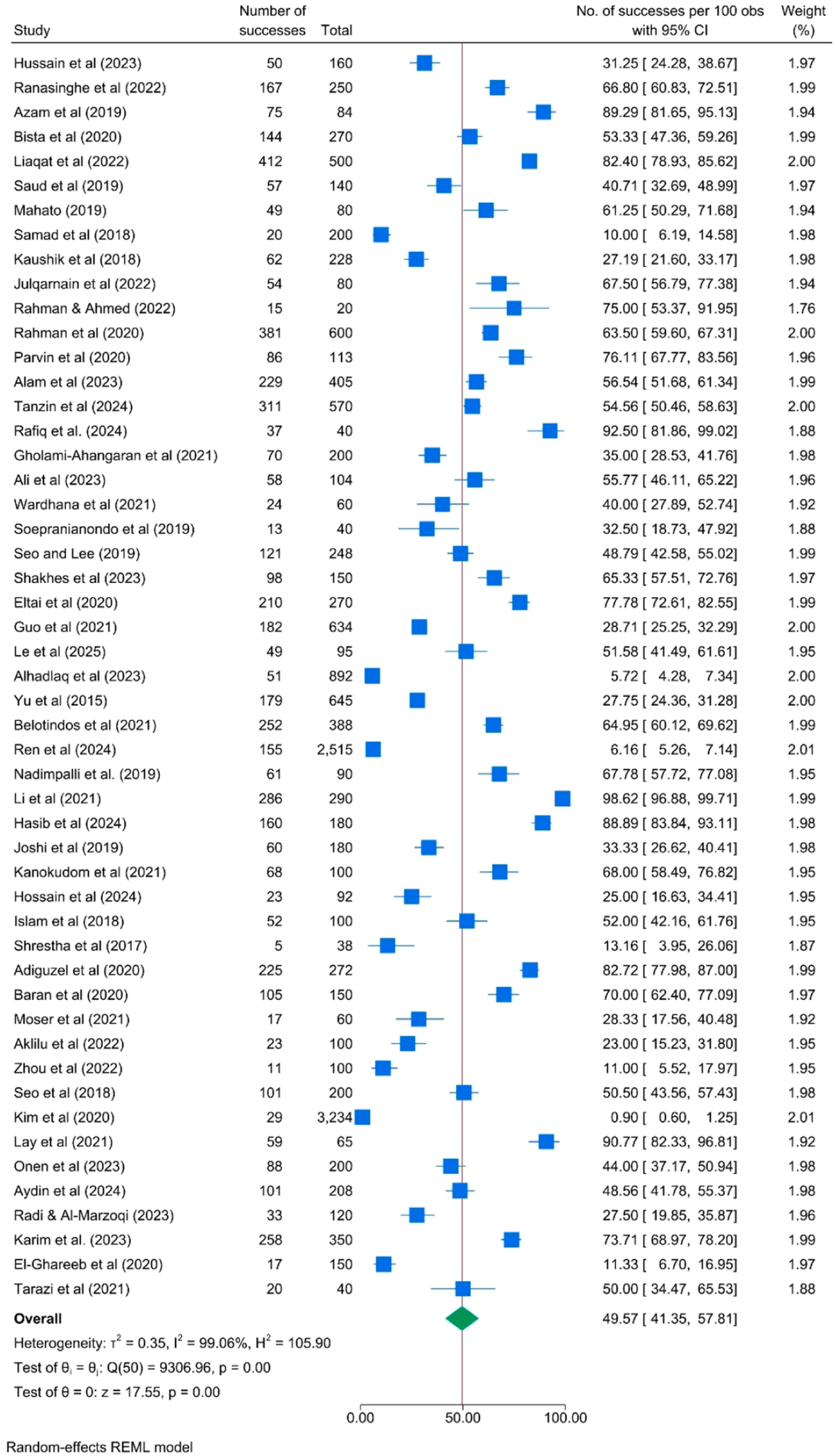


**Fig. 3.** Prevalence of *E. coli* in meat samples isolated from different species.

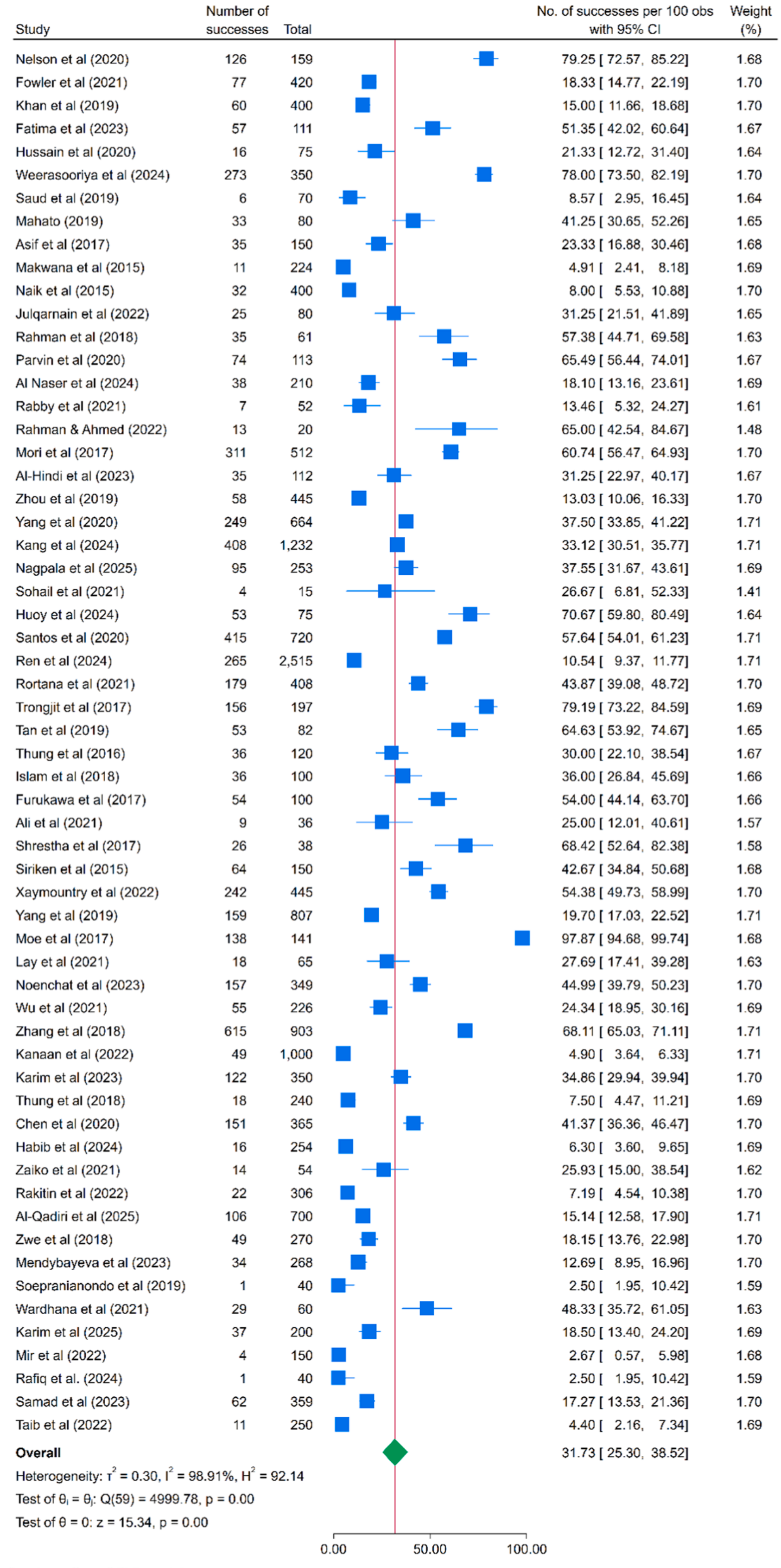


**Fig. 4.** Prevalence of *Salmonella* spp. in meat samples isolated from different species.

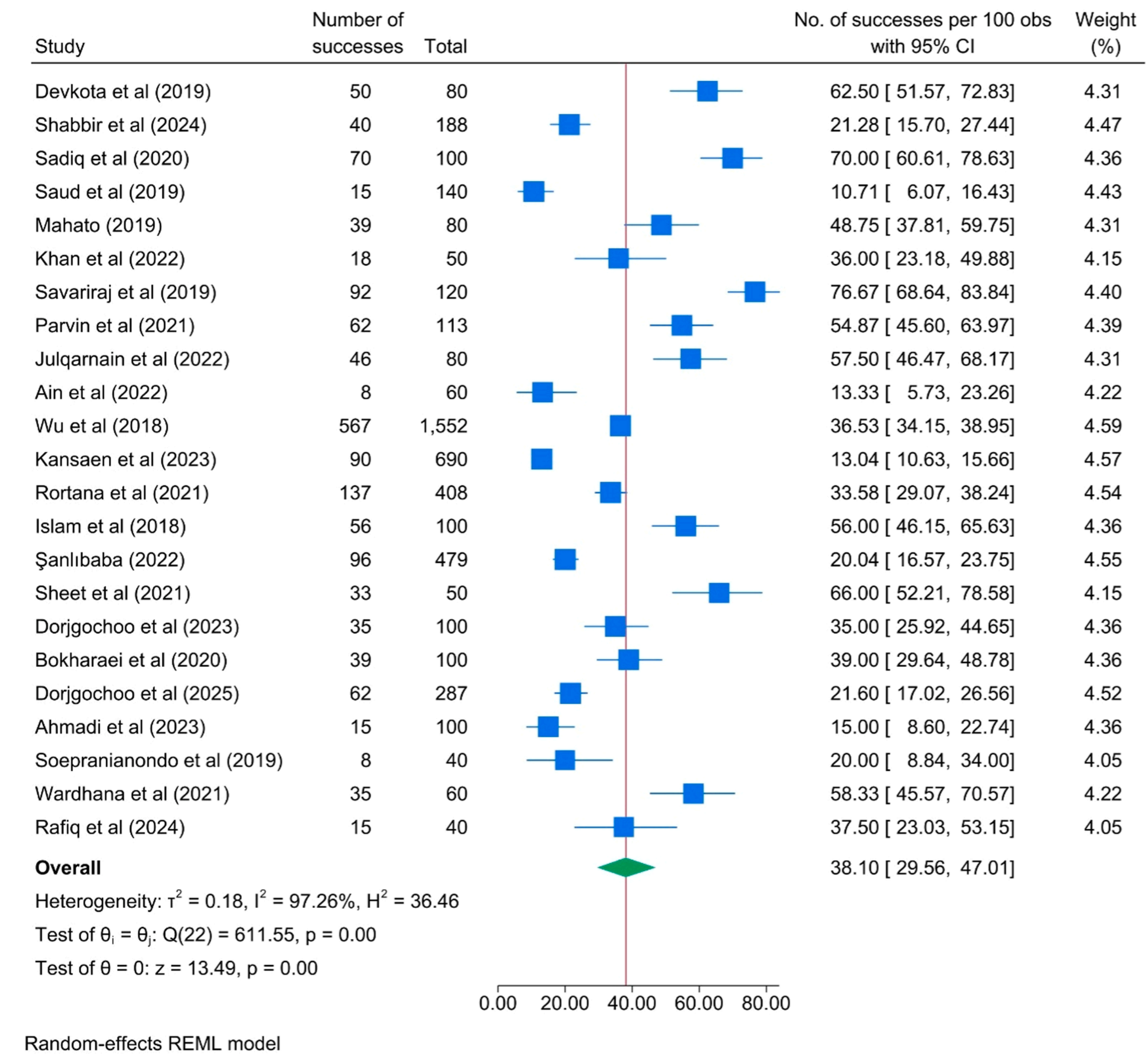


**Fig. 5.** Prevalence of *S. aureus* in meat samples isolated from different species.

amoxicillin (69.2 %, 95 % CI: 37.8–93.5 %), tetracycline (66.1 %, 95 % CI: 50.8–79.8 %), nalidixic acid (65.3 %, 95 % CI: 47.9–80.8 %), streptomycin (63.8 %, 95 % CI: 52.7–74.2 %), and TMP-SMX (61 %, 95 % CI: 45.4–75.6 %). Moderate resistance was observed for cefuroxime (53.6 %, 95 % CI: 10.9–93.3 %), doxycycline (49.6 %, 95 % CI: 31.6–67.6 %), ciprofloxacin (48.5 %, 95 % CI: 32.7–64.4 %), kanamycin (46.2 %, 95 % CI: 29.4–63.4 %), and chloramphenicol (40.6 %, 95 % CI: 29.8–52 %). Low resistance rates were reported for ceftriaxone (21.4 %, 95 % CI: 9.7–35.9 %), amikacin (19.4 %, 95 % CI: 4.9–39.7 %), cefoxitin (19 %, 95 % CI: 4.739.5 %), and colistin (18.6 %, 95 % CI: 11.3–27.3 %).

*Salmonella* spp. presented the greatest resistance to oxytetracycline (71.3 %, 95 % CI: 29.5–99.2 %), tetracycline (67.9 %, 95 % CI: 54–80.4 %) and erythromycin (65.9 %, 95 % CI: 22.9–98.1 %), with moderate resistance to TMP-SMX (41.2 %, 95 % CI: 28.3–54.7 %), nalidixic acid (45.8 %, 95 % CI: 31.7–60.2 %), and ampicillin (48.2 %, 95 % CI: 34.1–62.6 %). Resistance to third-generation cephalosporins such as cefotaxime (14.2 %, 95 % CI: 7.5–22.3 %) and ceftriaxone (13.1 %, 95 % CI: 4–25.5 %) was markedly lower.

In contrast, *S. aureus* isolates presented the highest resistance to ampicillin (84.7 %, 95 % CI: 67.4–96.7 %), followed by oxacillin (74.8 %, 95 % CI: 57.7–88.9 %), tetracycline (53.5 %, 95 % CI: 25.4–80.6 %), and ciprofloxacin (25 %, 95 % CI: 13.5–38.6 %). TMP-SMZ (5.1 %, 95 % CI: 3.7–6.6 %), gentamicin (19.1 %, 95 % CI: 9.1–31.5 %), and chloramphenicol (20.8 %, 95 % CI: 3.5–46.3 %) had relatively low resistance rates.

Among the three less frequently reported bacterial genera, *Klebsiella* isolates demonstrated very high resistance to amoxicillin (90.9 %, 95 % CI: 47.8–100 %) and ampicillin (97.7 %, 95 % CI: 61.7–100 %), although the data were limited to two studies each. Gentamicin showed much lower resistance in *Klebsiella* (4.1 %, 95 % CI: 0–30.6 %), while moderate resistance was found for nalidixic acid (30.2 %, 95 % CI: 0.1–75.9 %).

In the case of *Enterococcus* spp., resistance was most prominent to streptomycin (50 %, 95 % CI: 40.2–59.9 %) and tetracycline (51.4 %, 95 % CI: 41.9–60.7 %). Moderate resistance was also detected for ciprofloxacin (42.3 %, 95 % CI: 33.1–51.8 %) and erythromycin (37.3 %, 95 % CI: 3.9–80.1 %), whereas chloramphenicol resistance was relatively

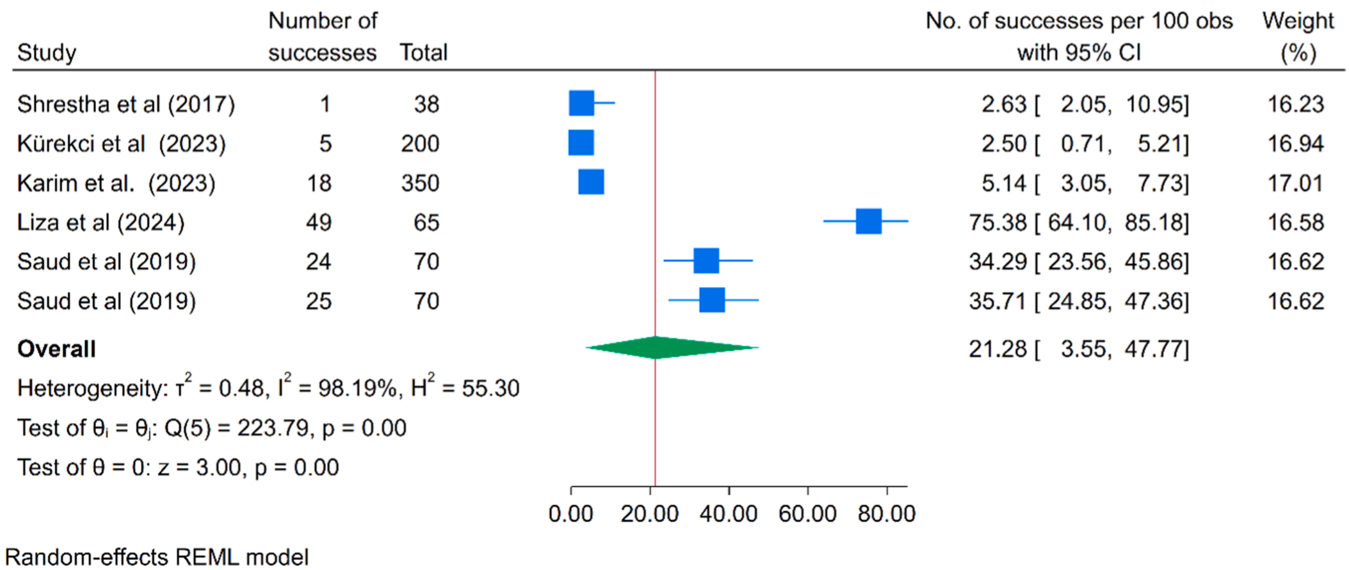


**Fig. 6.** Prevalence of *Klebsiella* spp. in meat samples isolated from different species.

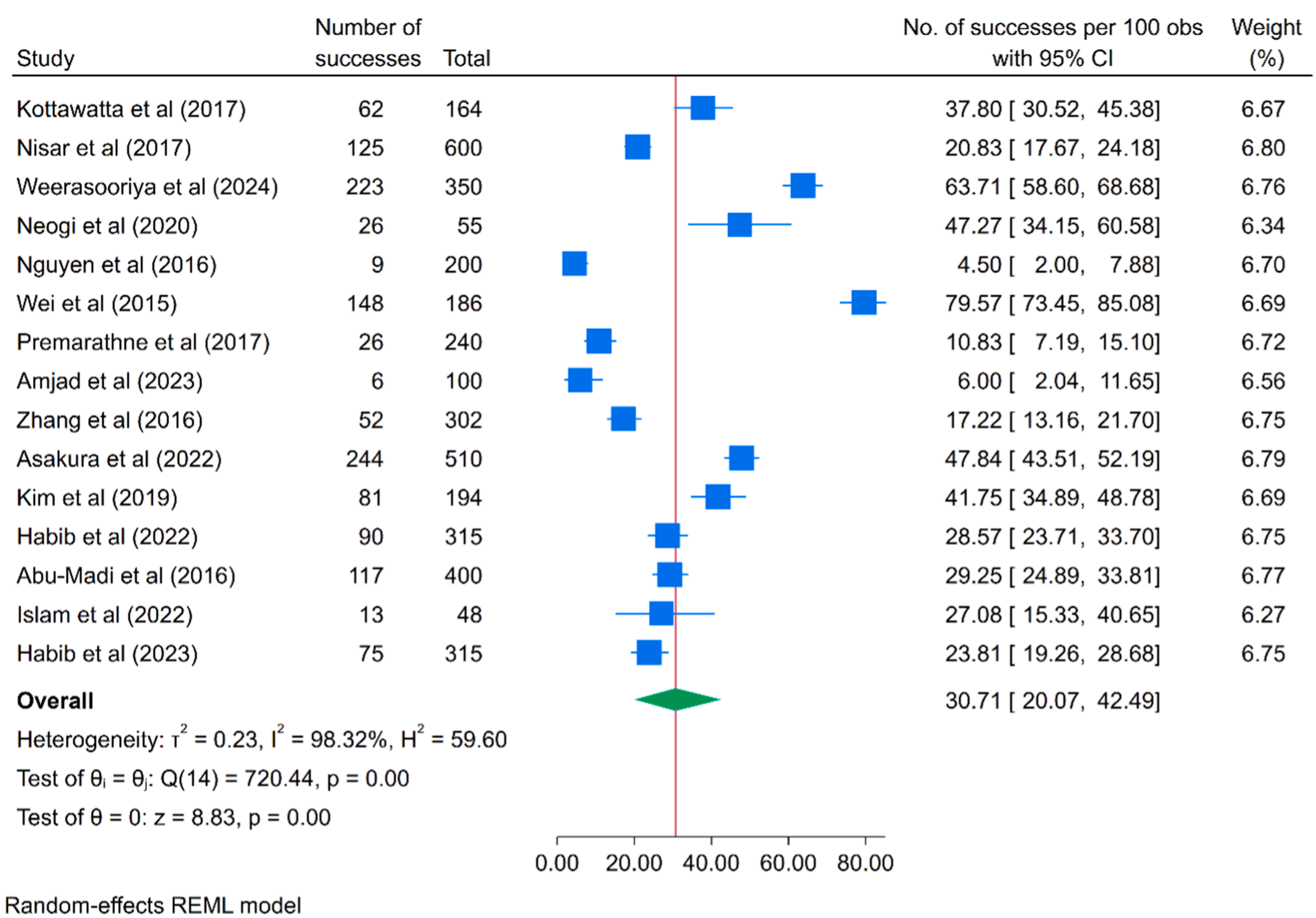


**Fig. 7.** Prevalence of *Campylobacter* spp. in meat samples isolated from different species.

low (4 %, 95 % CI: 0–15.8 %).

The *Campylobacter* isolates presented the most significant resistance to ciprofloxacin (75.5 %, 95 % CI: 59–89 %), followed by nalidixic acid (65.6 %, 95 % CI: 42.3–85.8 %) and tylosin (79.7 %, 95 % CI: 71.6–86.9 %). Moderate resistance was observed for tetracycline (50.4 %, 95 % CI: 28.7–72 %) and streptomycin (32 %, 95 % CI: 3.3–70.3 %), whereas low resistance was reported for gentamicin (15.8 %, 95 % CI: 6.2–28.3 %), neomycin (17.3 %, 95 % CI: 4.7–34.9 %), and doxycycline (21.5 %, 95 % CI: 0–69.7 %). The resistance to macrolides was substantially low, with 8.2 % (95 % CI: 4–13.4 %) for azithromycin and 6 % (95 % CI: 0.5–15.4 %) for erythromycin (Table 3)

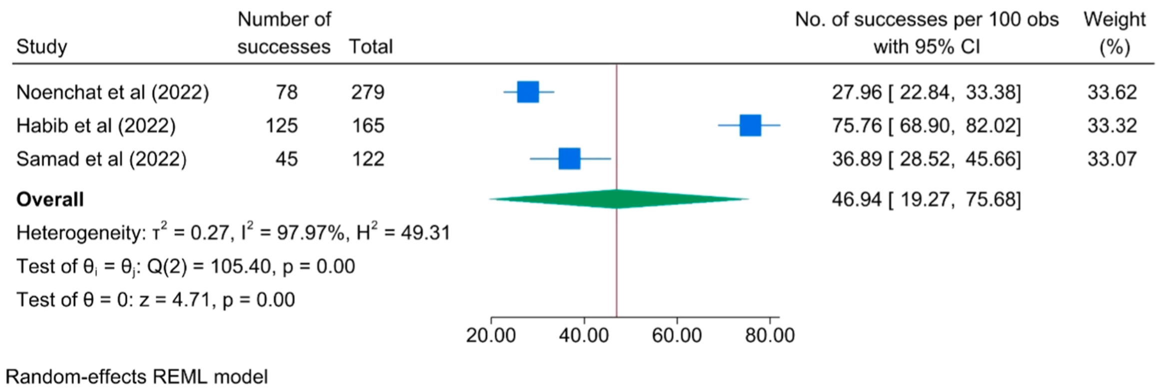


**Fig. 8.** Prevalence of *Enterococcus* spp. in meat samples isolated from different species.

### *Geographical AMR pattern across bacterial genera and antibiotic classes*

Owing to the limited number of studies available for certain bacterial species and antibiotic combinations, subgroup analysis could not be performed. Nevertheless, a heatmap depicting the distribution of antimicrobial resistance (AMR) across bacterial species in multiple Asian countries is shown as Fig. 16. Each cell represents the prevalence of a specific resistance trait within a bacterial species from a given country. The color gradient ranges from blue (low prevalence) to red (high prevalence), with intermediate values shown in purple. This visualization facilitates comparison of AMR patterns across countries and bacterial hosts. Comparative analysis of AMR across countries revealed varying resistance patterns for different antibiotic classes among the six studied bacterial genera. Bangladesh consistently presented the highest resistance levels to multiple antibiotic categories, including β-lactams, cephalosporins, fluoroquinolones, macrolides, phenicols, polymyxins, sulfonamides, and tetracyclines. Turkey presented the highest resistance rate to aminoglycosides and quinolones, followed by China and Bangladesh. Resistance to glycopeptides has been reported only in India. These findings highlight significant geographic differences in resistance trends, with Bangladesh emerging as a hotspot for multidrug resistance, particularly against broad-spectrum antibiotic classes.

### *Genotypic resistance pattern*

Although PCR has been used to detect resistance genes, determining the combined prevalence of genotypic AMR was challenging because of variations in the primers used across different studies. Therefore, the distribution of AMR genes reported across different studies is illustrated in Fig. 17. The heatmap provides a comparative overview, where each row corresponds to an individual study and each column represents a specific AMR gene. The color gradient reflects the prevalence of each gene, with red colors indicating higher prevalence and blue colors indicating lower prevalence. The β-lactam resistance genes detected included *blaCITM, blaCMY, blaCTX, blaDHA, blaIMP, blaNDM, blaOXA, blaSHV, blaTEM,* and *blaVEB*, indicating widespread resistance to both extended-spectrum and carbapenem antibiotics. Tetracycline resistance was conferred by *tetA, tetB, tetC, tetD*, and *tetO*, whereas sulfonamide resistance genes included *sul1, sul2,* and *sul3*. Fluoroquinolone resistance is predominantly mediated by *qnr, qnrA, qnrB*, and *qnrS*. Aminoglycoside resistance was represented by *aac (3), aac (6), aadA, aadA1, aph(3), aph(6),* and *strA*. Several trimethoprim resistance genes, such as *dfr1, dfr5, dfr7, dfr12,* and *dfrA, were also detected.* Macrolide resistance was linked to the presence of *ereA, ermC, mefA*, and *mefB*. Additionally, resistance to phenicols was associated with *cat1, cat2, catA1, cmlA*, and *floR*, whereas colistin resistance was mediated by *mcr-1* and *mcr-3*. The methicillin resistance genes *mecA* and *mecC* were also detected. Other notable ARGs included *int1* (integron class-1 integrase), *optrA* (oxazolidinone resistance), and the polymyxin resistance-associated genes *pmrC, pmrE*, and *pmrF*. Furthermore, *VIM-1*, a metallo-β-lactamase gene, was reported in some datasets. The prevalence of these AMR genes across the studies showed a wide range, ranging from 0.44 to 100 %. Significantly, *tetA, blaTEM, blaCTX, sul1, blaSHV,* and *mcr1* appeared to be the most prevalent ARGs across the studies.

## Discussion

To the best of our knowledge, this is the first systematic review and meta-analysis to estimate the pooled prevalence of six bacterial isolates (*E. coli, Salmonella* spp.*, S. aureus, Klebsiella* spp.*, Campylobacter* spp., and *Enterococcus* spp.) from the meat of various animal species, including cattle, goats, chickens, ducks, sheep, pigs, pigeons, buffaloes, turkeys, and camels, along with their AMR patterns to 26 antibiotics and associated genotypic profiles across twenty-eight (28) Asian countries. The data was collected through a systematic review of peer-reviewed publications from 2015 to 2025. This study provides essential insights into the prevalence of bacterial isolates (*E. coli, Salmonella* spp.*, S. aureus, Klebsiella* spp., *Campylobacter* spp., and *Enterococcus* spp.) from meat, along with their AMR profiles and associated genotypic characteristics.

In this meta-analysis, the pooled prevalence of *E. coli* in meat and meat products from 51 studies across Asian countries was 49.6 %, indicating significant heterogeneity ($I^2 = 99$ %, $p < 0.0001$). The findings of this study are compared with the outcomes of previous studies reported by Harun et al. (2024) (74 %, 95 % CI: 60–85.8 %) from livestock and poultry, by Dawadi et al. (2021) (83.5 %, 95 % CI: 65.1–93.2 %) from poultry, and by Zelalem et al. (2019) (5 %. 95 % CI: 4–7 %), and Islam et al. (2023) (69.3 %, 95 % CI: 67.3–71.2 %) from poultry and poultry environments. However, another study in China reported a lower prevalence of *E. coli* in food commodities, which was 4.3 % (3.3–5.2 %) (Paudyal et al., 2018). This review revealed a high prevalence of *E. coli* in poultry, which is usually part of the gut microbiota and may lead to cross-contamination. However, certain strains can cause colibacillosis, a disease affecting organs such as the liver, kidneys, and spleen, and contaminate beef through fecal contact (Aslam et al., 2003; Abalaka et al., 2017; Ievy et al., 2020; Dawadi et al., 2021).

The pooled prevalence of *Salmonella* spp. in this study was 31.73 % ($I^2 = 99$ %, $p < 0.0001$). Similar findings were reported by Harun et al. (2024) in South Asia and Ripon et al. (2023) in Bangladesh; the pooled prevalence of *Salmonella* spp. was approximately 37 %. However, marked variability was observed across regions, with lower estimates reported in China, Europe, and Africa (1.9–19.7 %) (Xavier et al., 2014; Gonçalves-Tenório et al., 2018; Yang et al., 2019; Zelalem et al., 2019), and a higher rate was detected in Pakistan at 51.4 % (Fatima et al., 2023). The presence of *Salmonella* spp. in meat and meat products is due to their spread through fecal contamination during slaughter and cross-contamination from undercooked meat to raw foods, and their environmental adaptability and ability to utilize diverse nutrients help them survive in meat products despite processing (Chagas et al., 2017;

**Table 1**
Subgroup meta-analysis of overall isolated bacteria regardless of the type of identified organism.

| Subgroup | Name/ Variables | % (95 % CI) | $\tau^2$ | $I^2$ | p value |
|---|---|---|---|---|---|
| Country | Afghanistan | 37.84 [12.25, 67.56] | 0.34 | 95.20 | <0.0001 |
| | Bangladesh | 39.20 [31.25, 47.44] | 0.25 | 96.51 | <0.0001 |
| | Cambodia | 57.17 [43.61, 70.22] | 0.16 | 95.27 | <0.0001 |
| | China | 26.97 [19.41, 35.24] | 0.28 | 98.89 | <0.0001 |
| | India | 20.88 [6.84, 39.65] | 0.33 | 97.61 | <0.0001 |
| | Indonesia | 31.75 [15.19, 50.99] | 0.21 | 91.36 | <0.0001 |
| | Iran | 22.27 [2.62, 52.41] | 0.30 | 97.78 | 0.01 |
| | Iraq | 21.11 [1.73, 52.99] | 0.46 | 99.02 | 0.01 |
| | Japan | 36.95 [16.06, 59.97] | 0.27 | 98.82 | <0.0001 |
| | Nepal | 35.58 [25.77, 46.03] | 0.23 | 95.39 | <0.0001 |
| | Myanmar | 77.76 [26.07, 100.00] | 0.81 | 98.55 | <0.0001 |
| | Jordan | 30.27 [3.70, 67.56] | 0.28 | 95.59 | 0.01 |
| | Kazakhstan | 12.69 [8.95, 16.96] | 0.00 | 0.00 | - |
| | Laos | 41.26 [21.69, 62.35] | 0.22 | 96.08 | <0.0001 |
| | Malaysia | 28.59 [12.29, 48.41] | 0.34 | 98.70 | <0.0001 |
| | Mongolia | 25.15 [14.74, 37.22] | 0.04 | 85.03 | <0.0001 |
| | Pakistan | 32.82 [20.44, 46.48] | 0.41 | 98.25 | <0.0001 |
| | Philippines | 62.05 [49.52, 73.83] | 0.11 | 95.02 | <0.0001 |
| | Qatar | 53.97 [10.29, 94.05] | 0.51 | 99.40 | <0.0001 |
| | Russia | 9.64 [4.93, 15.52] | 0.03 | 61.45 | <0.0001 |
| | Saudi Arab | 9.77 [2.90, 19.83] | 0.10 | 94.95 | <0.0001 |
| | Singapore | 24.19 [8.69, 44.31] | 0.18 | 97.65 | <0.0001 |
| | South Korea | 23.08 [11.60, 37.02] | 0.35 | 99.19 | <0.0001 |
| | Sri Lanka | 58.78 [41.46, 75.06] | 0.15 | 96.85 | <0.0001 |
| | Thailand | 28.69 [12.72, 47.99] | 0.24 | 98.17 | <0.0001 |
| | Turkey | 39.75 [20.21, 61.13] | 0.47 | 98.71 | <0.0001 |
| | UAE | 31.44 [6.96, 63.52] | 0.44 | 99.12 | <0.0001 |
| | Vietnam | 23.39 [2.53, 55.33] | 0.43 | 96.58 | 0.01 |
| Source | Farm | 37.49 [12.92, 66.04] | 0.67 | 98.73 | <0.0001 |
| | Market | 33.12 [29.40, 36.94] | 0.31 | 98.43 | <0.0001 |
| | Port custom | 5.47 [4.02, 7.11] | 0.00 | 0.01 | - |
| | Processing Plant | 40.34 [20.85, 61.50] | 0.30 | 98.11 | <0.0001 |
| | Restaurant | 14.62 [8.05, 22.54] | 0.00 | 0.00 | - |
| | Slaughterhouse | 37.26 [27.16, 47.93] | 0.33 | 97.53 | <0.0001 |
| Species | Buffalo | 24.26 [12.07, 38.80] | 0.14 | 86.82 | <0.0001 |
| | Camel | 11.33 [6.70, 16.95] | 0.00 | 0.00 | - |
| | Cattle | 27.83 [20.39, 35.90] | 0.29 | 98.12 | <0.0001 |
| | Chicken | 37.56 [32.87, 42.37] | 0.33 | 98.42 | <0.0001 |
| | Duck | 33.45 [19.20, 49.40] | 0.21 | 97.07 | <0.0001 |
| | Goat | 15.65 [3.21, 34.06] | 0.20 | 95.13 | <0.0001 |
| | Mix | 34.36 [2.34, 78.47] | 0.63 | 97.67 | 0.01 |
| | Pig | 35.18 [24.44, 46.73] | 0.37 | 99.03 | <0.0001 |
| | Pigeon | 39.29 [21.83, 58.17] | 0.00 | 0.00 | - |
| | Sheep | 17.22 [8.53, 28.03] | 0.22 | 97.17 | <0.0001 |
| | Turkey | 35 [28.53, 41.76] | 0.00 | 0.00 | - |
| Sample | Meat | 33.57 [30.02, 37.20] | 0.33 | 98.42 | <0.0001 |
| | Meat product | 29.45 [17.89, 42.48] | 0.26 | 97.90 | <0.0001 |

Stojiljkovic, 2018). Approximately 1 in every 25 chicken packages sold in grocery stores is contaminated with *Salmonella* (Brown et al., 2017).

This study revealed that the pooled prevalence of *S. aureus* was 38.1 % ($I^2 = 97$ %, $p < 0.0001$), which is lower than that reported in a meta-analysis based on South Asia (48 %) by Harun et al. (2024). The current findings are also compared with other studies where the prevalence of *S. aureus* was varied widely, from less than 5 % in Nigeria and South America (Khanal et al., 2022; Odetokun et al., 2023) to more than 70 % in chicken meat found in Nigeria (Odetokun et al., 2023). These discrepancies may arise from differences in meat type, hygiene practices, and detection methodologies. Moreover, *S. aureus* can enter meat from animal skin, intestines, or nasal passages during slaughter. It may also be introduced by human handlers or persist in processing environments due to poor hygiene and biofilm formation (Thwala et al., 2021; Pérez-Boto et al., 2023).

In this findings, *Klebsiella* spp. had a pooled prevalence of 21.3 % from meat and meat products, similar to the findings of another study (22.1 %) conducted by Muinde et al. (2023) in Kenya. Other studies revealed a relatively high percentage of Klebsiella spp., including 28.9 % in Bangladesh (Liza et al., 2024) and 47 % in meat products. The contamination is varied by meat type, including pork (58 %), chicken (47 %), and turkey (38 %) (Davis et al., 2015). Moreover, the highest prevalence was observed in Greece (81.8 %) (Theocharidi et al., 2022). The detection of *Klebsiella* spp. in the food chain often indicates poor hygiene during food preparation, handling, or storage, and its presence in ready-to-eat cooked food suggests undercooking or contamination post-cooking (Theocharidi et al., 2022).

This meta-analysis reported a pooled prevalence of 30.7 % for *Campylobacter* spp. A few meta-analyses conducted in Iran reported a pooled prevalence of 43.9 % in white meat and 7.9 % in red meat (Ansarifar et al., 2023). An analysis of 31 studies revealed an overall pooled prevalence of 23.4 % for *Campylobacter* spp. in meat and meat products, where duck meat presented the highest prevalence at 70.5 %, followed by chicken at 36.2 %, pork at 2.1 %, and beef at 1 % (Je et al., 2023). The pooled prevalence of *Enterococcus* spp. in this study was 46.9 %, whereas a study from the USA reported a considerably higher isolation rate of 92 % from retail meat samples, including chicken, turkey, beef, and pork (Tyson et al., 2018).

Univariate analysis revealed that factors such as sample type, source, and country may influence *E. coli* prevalence. In contrast, variations in sample type and study year appear to affect *Salmonella* and *S. aureus* prevalence. For *Klebsiella*, country, species, and source were potential contributors, whereas source and country were relevant for *Enterococcus* spp. and source alone for *Campylobacter* spp. The differences between this meta-analysis and studies from other regions may stem from diverse hygiene practices and environmental conditions across geographical locations and sample sources (Essack, 2021; Harun et al., 2024).

The overuse of antimicrobials in humans and animals leads to the emergence of MDR pathogens and reduced treatment effectiveness (Muinde et al., 2023). In this review, ampicillin, oxytetracycline, and tetracycline emerged as the antibiotics with the highest resistance levels across the investigated bacterial isolates, followed by ampicillin and nalidixic acid. Specifically, *E. coli and Salmonella* spp. presented the most significant resistance to oxytetracycline, an antibiotic of the tetracycline class. Tetracyclines represent the most extensively used class of antimicrobials globally, with an estimated consumption of 33,305 tons, and

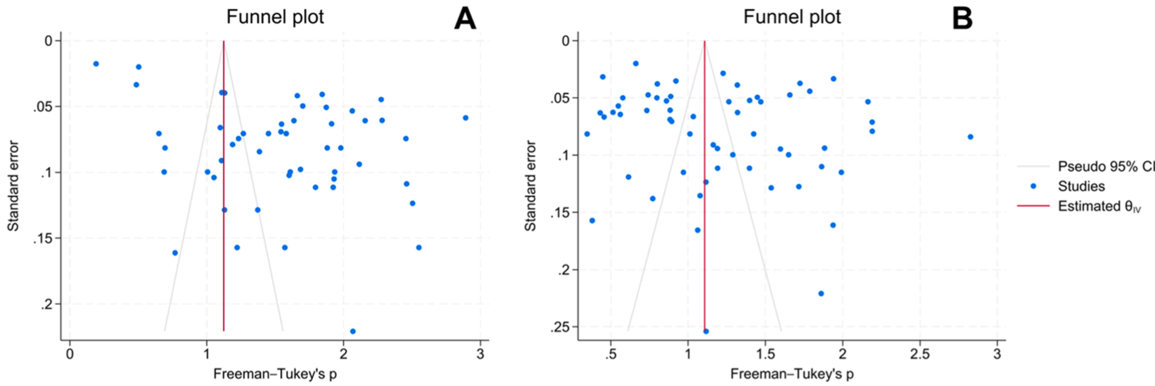


**Fig. 9.** Funnel plot [A] *E. coli* [B] *Salmonella* spp. for assessing study bias.

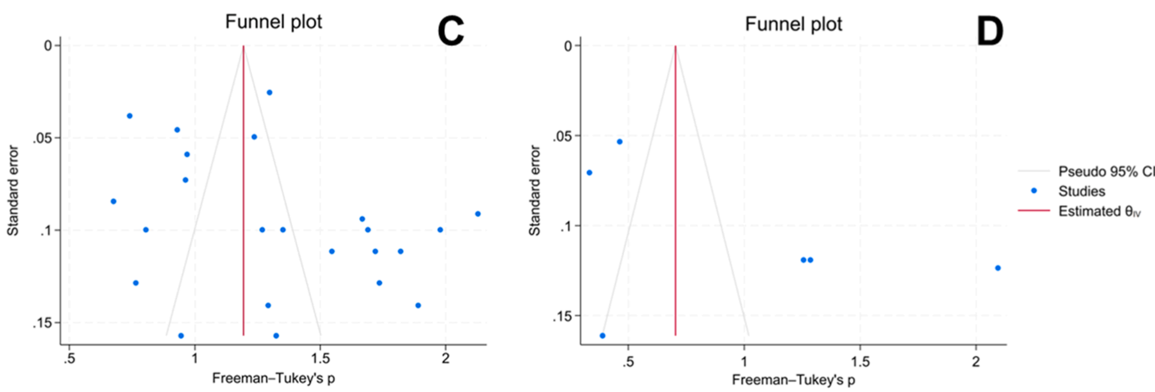


**Fig. 10.** Funnel plot [C] of *S. aureus* [D] *Klebsiella* spp. for assessing study bias.

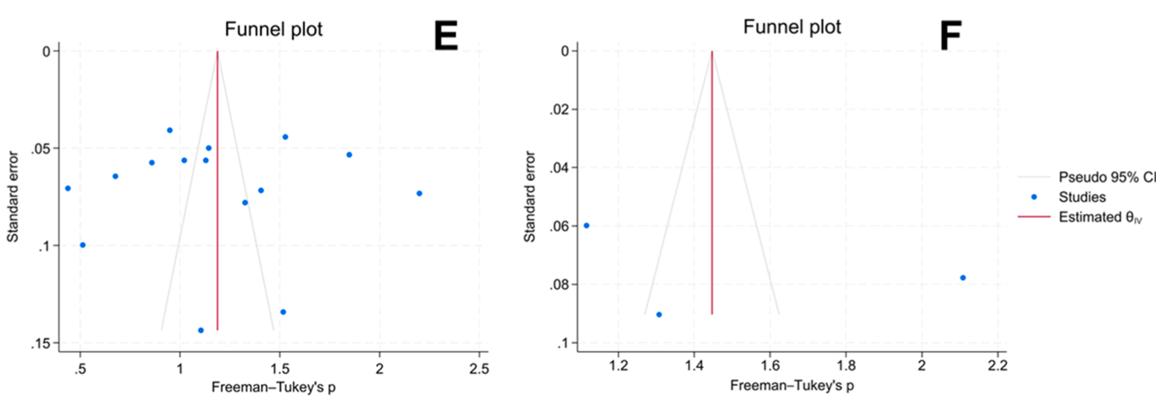


**Fig. 11.** Funnel plot [E] *Campylobacter* spp. [F] *Enterococcus* spp. for assessing study bias.

are projected to increase by 9 % by 2030. Nonetheless, the intensity of antimicrobial use (AMU) by class varies across countries; for example, penicillin use is highest in Thailand, whereas Chile has reported the highest prevalence of amphenicol usage (Mulchandani et al., 2023).

The most AMU intensity hotspots, approximately 67 %, were concentrated in Asia. The primary regions included eastern China, southern India, Central Java (Indonesia), central Thailand, the eastern coastline of Vietnam, western South Korea, eastern India, Bangladesh, Pakistan, and northwestern Iran (Mulchandani et al., 2023). Moreover, veterinarians commonly use tetracycline and aminoglycosides for domestic animals (Khan et al., 2020). Other antibiotics in the tetracycline class, including tetracycline and doxycycline, and beta-lactams showed significant resistance against *E. coli, Salmonella* spp., *Enterococcus* spp., *Campylobacter* spp., and *S. aureus*. This widespread resistance could be attributed to the extensive and prolonged use of tetracyclines and beta-lactams in veterinary practice as growth promoters and prophylactics (Peng et al., 2014). All the genera of the isolated bacteria presented significant resistance to ampicillin and amoxicillin. This is because ampicillin and amoxicillin have been among the most commonly prescribed β-lactam antibiotics worldwide for decades and are used therapeutically in humans and sub-therapeutically in animal feeds to promote growth and prevent disease (Rajaei et al., 2021).

*E. coli, Salmonella* spp., *Campylobacter* spp., and *S. aureus*, which were 66 %, 68 %, 66 %, and 54 %, respectively, presented significant resistance to nalidixic acid. This finding contradicts the study conducted by Harun et al. (2024), who reported that resistance to this antibiotic was not prominent. However, the outcome supports other studies reported by Baz et al. (2021) and Denton et al. (2008). Resistance to erythromycin and streptomycin was remarkable in *E. coli, Salmonella* spp., and *S. aureus*, while resistance to gentamicin and ciprofloxacin was one of

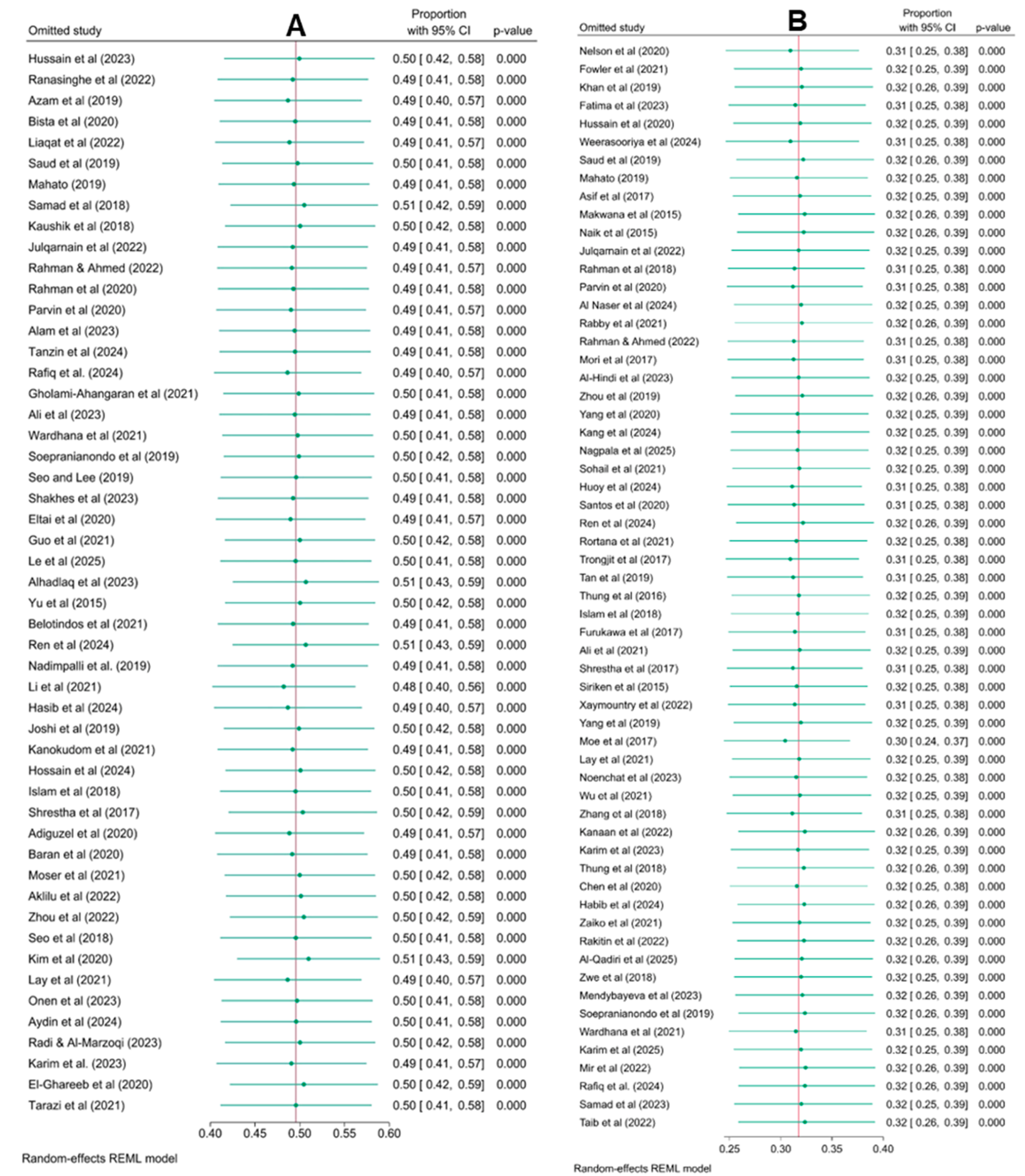


**Fig. 12.** Leave one study out of [A] *E. coli* [B] *Salmonella* spp.

the lowest in this study. However, only *Campylobacter* spp. were resistant (75.5 %) to ciprofloxacin in the present study. *Salmonella* spp. (61.3 %) were more resistant to colistin than were *E. coli* (18.6 %). Compared with the results of the study conducted by Harun et al. (2024), the prevalence was greater for *Salmonella* spp. (50.3 %), whereas it was lower for *E. coli* (28.3 %). The use of colistin, a last-resort antibiotic critical for treating MDR infections in humans, has been banned or restricted in several countries, such as India (Harun et al., 2024), Bangladesh (Tasmim et al., 2023), Oman (Ahmed et al., 2025), China (Olaitan et al., 2021), Nepal (Acharya and Wilson, 2019), and Brazil (Monte et al., 2017), owing to its role in promoting AMR when it is used in food-producing animals.

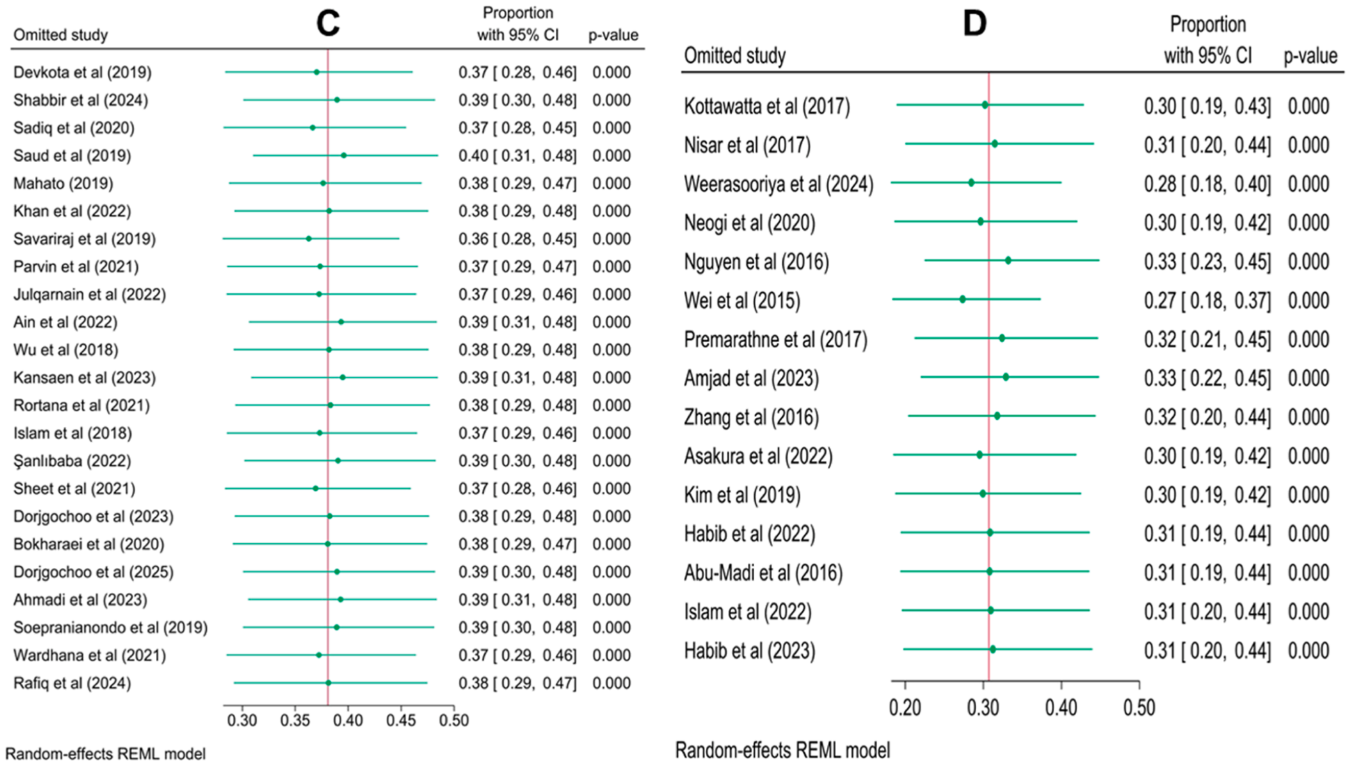


**Fig. 13.** Leave one study out of [C] *S. aureus* [D] *Campylobacter* spp.

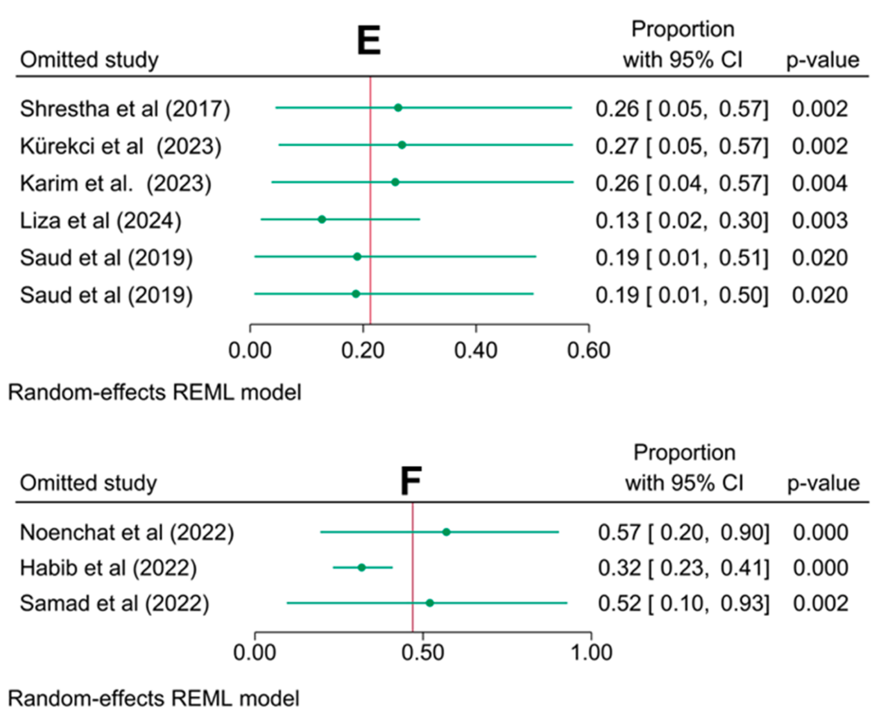


**Fig. 14.** Leave one study out of [E] *Klebsiella* spp. [F] *Enterococcus* spp.

The continuous and persistent contact between humans and livestock and poultry in farming systems has been recognized as a significant way for the transmission and dissemination of AMR bacteria (Matheou et al., 2025). Additionally, AMR can spread from livestock to humans through direct contact, placing farmers, veterinarians, and meat processing workers at increased risk of acquiring resistant infections because of their close and frequent exposure to animals and animal products (Neyra et al., 2012; Van Gompel et al., 2019). Resistant bacteria can also be transmitted through contact with animal waste, which is often used as fertilizer in agricultural fields. Manure, which contains both antibiotic residues and resistant bacteria, is applied to crops as a source of nutrients, but this practice inadvertently introduces resistance genes into soil and water systems, creating reservoirs of resistance that can affect wildlife and humans (Doyle et al., 2025). Over time, these

**Table 2**
Univariate meta-regression of covariates.

| Covariates | *E. coli* | | | | *Salmonella* spp. | | | | *S. aureus* | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | *β* | 95 % CI | *p* value | $R^2$(%) | *β* | 95 % CI | *p* value | $R^2$(%) | *β* | 95 % CI | *p* value | $R^2$(%) |
| Study year | 0.0161 | −0.0175 to 0.0497 | *0.348* | Nil | −0.0161 | −0.0385 to 0.0063 | 0.159 | 1.70 | −0.0436 | −0.0824 to −0.0047 | ***0.028*** | 15.91 |
| Country | −0.0033 | −0.0134 to 0.0069 | *0.531* | Nil | 0.0025 | −0.0064 to 0.0115 | 0.579 | Nil | −0.0069 | −0.0296 to 0.0157 | *0.547* | Nil |
| Species | 0.0216 | −0.0642 to 0.1075 | *0.621* | Nil | −0.0070 | −0.0299 to 0.0159 | 0.547 | Nil | 0.0263 | −0.0107 to 0.0632 | *0.163* | 4.36 |
| Sample | −0.1029 | −0.2181 to 0.0123 | *0.08* | 4.04 | −0.0364 | −0.1004 to 0.0276 | 0.265 | 0.57 | −0.1525 | −0.2898 to −0.0151 | ***0.029*** | 14.63 |
| Source | −0.0731 | −0.2012 to 0.0551 | *0.264* | 0.43 | 0.0313 | −0.0490 to 0.1117 | 0.445 | Nil | 0.1350 | −0.0269 to 0.2969 | *0.102* | 8.01 |
| Sample Size | −0.0002 | −0.0003 to −0.0001 | ***0.004*** | 13.09 | −0.0001 | −0.0002 to 0.0001 | 0.305 | Nil | −0.0001 | −0.0004 to 0.0001 | *0.284* | 0.50 |
| Positive Cases | 0.0012 | 0.0005 to 0.0018 | ***0.0005*** | 18.90 | 0.0009 | 0.0004 to 0.0013 | **0.0003** | 18.26 | 0.0001 | −0.0007 to 0.0008 | *0.879* | Nil |
| Covariates | ***Klebsiella* spp.** | | | | ***Enterococcus* spp.** | | | | ***Campylobacter* spp.** | | | |
| | ***β*** | **95 % CI** | ***p* value** | **$R^2$(%)** | ***β*** | **95 % CI** | ***p* value** | **$R^2$(%)** | ***β*** | **95 % CI** | ***p* value** | **$R^2$(%)** |
| Study year | 0.0519 | −0.1833 to 0.2871 | *0.665* | Nil | - | - | - | - | 0.0112 | −0.0715 to 0.0939 | *0.791* | Nil |
| Country | −0.4420 | −0.9438 to 0.0597 | *0.084* | 28.73 | 0.1950 | −0.1275 to 0.5176 | 0.2359 | 17.36 | 0.0013 | −0.0874 to 0.0899 | *0.977* | Nil |
| Species | −0.5021 | −1.1561 to 0.1518 | *0.132* | 20.84 | −0.1493 | −0.9630 to 0.6643 | 0.7191 | Nil | 0.0778 | −0.2279 to 0.3835 | *0.618* | Nil |
| Sample | 1.3522 | 0.3147 to 2.3897 | ***0.011*** | 53.95 | −0.4400 | −0.5884 to −0.2916 | Nill | 95.81 | −0.0628 | −0.5002 to 0.3747 | *0.779* | Nil |
| Source | 0.0391 | −1.0442 to 1.1222 | *0.944* | Nil | −0.2847 | −0.9422 to 0.3728 | 0.3961 | Nil | −0.2888 | −0.6163 to 0.0388 | *0.084* | 12.35 |
| Sample Size | −0.0029 | −0.0077 to 0.0019 | *0.236* | 8.23 | −0.0013 | −0.0069 to 0.00425 | 0.6422 | Nil | 0.0001 | −0.0015 to 0.0017 | *0.882* | Nil |
| Positive Cases | 0.0392 | 0.0253 to 0.0531 | ***0.000*** | 87.79 | 0.0052 | −0.0018 to 0.0123 | 0.1436 | 37.08 | 0.0044 | 0.0018 to 0.0070 | ***0.001*** | 43.30 |

resistant bacteria can be carried into surface water through runoff or leach into groundwater, further spreading AMR in the environment (Kusi et al., 2022; Yang et al., 2025).

The findings of this study revealed that *tetA*, which encodes an efflux pump conferring resistance to tetracyclines, was the most frequently detected resistance gene. sul1, which is responsible for sulfonamide resistance, is commonly found alongside other resistance genes on integrons and promotes the horizontal transfer of genetic material between bacterial populations (Naas et al., 2006; Koczura et al., 2016; Poey et al., 2019). Moreover, the β-lactamase-encoding genes *blaTEM, blaSHV*, and *blaCTX* are related to resistance to β-lactams (penicillin and cephalosporins), and *blaCTX* is commonly detected in *Enterobacteriaceae* isolated from meat and meat products (Hayashi et al., 2018). Furthermore, a major concern is the detection of *mcr-1*, a gene that confers resistance to colistin, a last-resort antibiotic used to treat MDR Gram-negative infections in humans (Mondal et al., 2024).

The most frequent detection of *sul1, blaTEM,* and *blaCTX* suggests their localization within class 1 integrons, which are known to capture and disseminate multiple resistance cassettes simultaneously (Li et al., 2016). Similarly, co-detection of *mcr-1* with *blaCTX-M* and *blaTEM* on conjugative plasmids (IncF, IncHI2, IncI2) has been reported across studies, indicating the presence of mobile platforms capable of horizontal gene transfer (HGT) among bacterial populations (Liu et al., 2020; Che et al., 2021; Feng et al., 2023). These plasmids not only enable interspecies transfer but also enhance the persistence of resistance determinants in diverse ecological niches (Bottery, 2022). Our findings, therefore, corroborate growing evidence that plasmid-mediated HGT is a pivotal mechanism driving the dissemination of ARGs. Class 1 integrons identified in *E. coli* isolates from cooked meat have been demonstrated to mobilize resistance cassettes through both conjugation and transformation (Yu et al., 2016). This implies that resistant bacteria from food sources can potentially transmit ARGs to commensal or pathogenic bacteria within the human gut, amplifying the risk of treatment failures and limiting therapeutic options (Paul and Das, 2022).

In our findings, genes such *as tetA, sul1, blaTEM, blaCTX, mcr-1,* and *mcr-3* were among the most frequently detected resistance determinants, reflecting their central roles in conferring phenotypic resistance to tetracyclines, sulfonamides, β-lactams, and colistin. The co-occurrence of *mcr-1* with extended-spectrum β-lactamase (ESBL) genes such as *blaCTX-M* and *blaTEM* in *E. coli* isolates has been repeatedly linked with multidrug-resistant profiles, heightening both clinical and public health concerns (Zheng et al., 2020). The first report showing the co-carriage of *mcr-1* with *blaCTX-M-1, blaTEM,* and *blaSHV* in *E. coli* from chickens afflicted with colibacillosis underscores the risk of widespread resistance dissemination across food-producing animals (Dhaouadi et al., 2020). Moreover, epidemiological surveillance reveals that tetA was detected in 67 % of tetracycline-resistant E. coli isolates, while *blaTEM* and *blaCTX* were present in 36 % and 18 % of β-lactam-resistant isolates, respectively (Wang et al., 2022).

In agriculture, workers such as farmers and veterinarians who have frequent contact with livestock are at heightened risk of contracting infections caused by resistant bacteria. In aquaculture, the addition of antibiotics to water can lead to the release of resistant bacteria into nearby water bodies, affecting wild aquatic species and contributing to environmental AMR reservoirs (Martinez, 2009). In terrestrial systems, the use of animal waste as fertilizer introduces antibiotic-resistant bacteria and resistance genes into soil, which can spread to water bodies via runoff, impacting water quality and public health (Kümmerer, 2009). Global trade, travel, and animal migration further facilitate the spread of AMR, with examples such as AMR *E. coli* detected in imported frozen shrimp in Saudi Arabia, illustrating the risks of international trade (Alhabib and Elhadi, 2024).

There has been no meta-analysis conducted on collective bacterial resistance to multiple antibiotics (28 antibiotics) in Asian countries; the outcomes of this study can be compared with those of a previous meta-analysis by Harun et al. (2024), which focused on only three (3) bacterial genera, while six (6) genera were included in the present study.

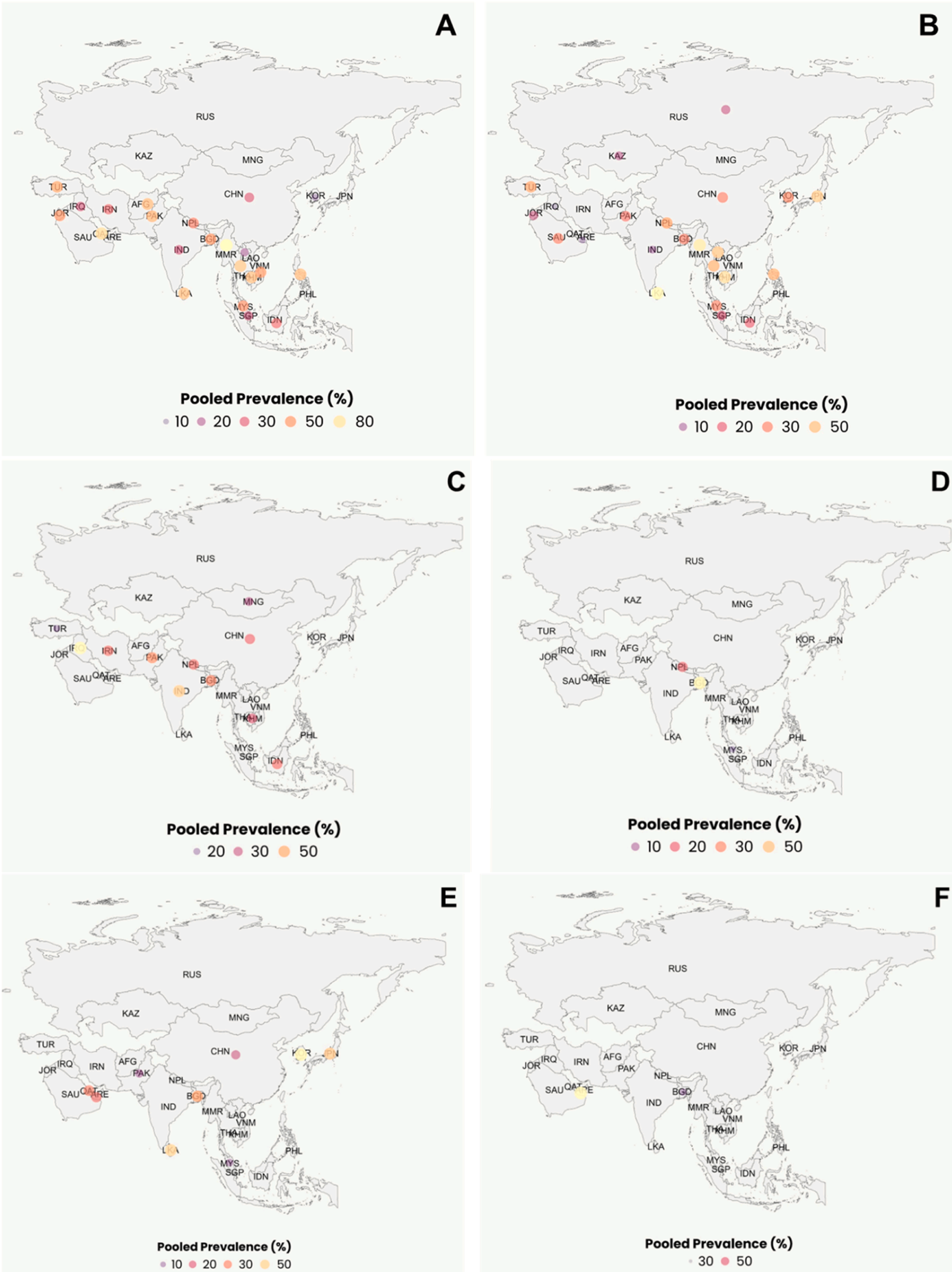


**Fig. 15.** Pooled prevalence of isolated bacteria [A] *E. coli*; [B] *Salmonella* spp.; [C] *S. aureus*; [D] *Klebsiella* spp.; [E] *Campylobacter* spp.; [F] *Enterococcus* spp. The figure was created via R programming.

**Table 3**
Overall meta-analysis of AMR carriage.

| Antibiotics | *E. coli* | | | | *Salmonella* spp. | | | | *S. aureus* | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | N | % (95 % CI*) | $\tau^2$ | $I^2$ | N | % (95 % CI*) | $\tau^2$ | $I^2$ | N | % (95 % CI*) | $\tau^2$ | $I^2$ |
| Amoxicillin | 7 | 69.20 [37.75, 93.45] | 0.67 | 96.7 | 12 | 39.39 [20.12, 60.39] | 0.47 | 95.0 | 5 | 60.45 [21.68, 93.07] | 0.77 | 99.8 |
| Ampicillin | 15 | 79.67 [66.65, 90.23] | 0.31 | 96.2 | 24 | 48.24 [34.08, 62.55] | 0.47 | 97.8 | 8 | 84.73 [67.36, 96.66] | 0.30 | 96.3 |
| Azithromycin | 3 | 37.46 [17.90, 59.26] | 0.11 | 82.7 | 10 | 48.76 [24.15, 73.67] | 0.62 | 96.7 | 2 | 44.29 [36.57, 52.15] | 0.00 | 0.00 |
| Cefotaxime | 8 | 29.90 [13.55, 49.32] | 0.30 | 96.2 | 13 | 14.15 [7.50, 22.30] | 0.12 | 93.8 | - | - | - | - |
| Ceftriaxone | 7 | 21.39 [9.65, 35.93] | 0.15 | 90.0 | 10 | 13.08 [3.98, 25.53] | 0.20 | 94.12 | - | - | - | - |
| Chloramphenicol | 12 | 40.64 [29.77, 51.97] | 0.14 | 91.8 | 18 | 33.02 [22.39, 44.55] | 0.23 | 96.2 | 8 | 20.84 [3.52, 46.30] | 0.57 | 97.8 |
| Ciprofloxacin | 16 | 48.48 [32.69, 64.43] | 0.39 | 96.9 | 20 | 27.13 [16.54, 39.09] | 0.28 | 96.33 | 6 | 25.04 [13.49, 38.63] | 0.11 | 91.9 |
| Erythromycin | 3 | 57.99 [32.70, 81.32] | 0.17 | 88.3 | 7 | 65.87 [22.87, 98.06] | 1.16 | 93.8 | 6 | 45.60 [28.81, 62.90] | 0.16 | 93.3 |
| Streptomycin | 7 | 63.77 [52.69, 74.20] | 0.07 | 82.13 | 18 | 46.62 [35.48, 57.93] | 0.20 | 95.3 | 3 | 41.63 [10.43, 77.02] | 0.39 | 97.6 |
| Tetracycline | 16 | 66.05 [50.80, 79.82] | 0.37 | 96.9 | 23 | 67.86 [53.95, 80.42] | 0.42 | 97.6 | 6 | 53.53 [25.35, 80.59] | 0.49 | 98.1 |
| TMP-SMZ* | 11 | 61.02 [45.42, 75.58] | 0.25 | 95.4 | 19 | 41.19 [28.30, 54.66] | 0.29 | 95.3 | 3 | 5.07 [3.69, 6.63] | 0.00 | 0.00 |
| Gentamicin | 17 | 26.07 [18.7, 34.91] | 0.14 | 91.2 | 17 | 16.86 [11.91, 22.41] | 0.06 | 86.8 | 5 | 19.07 [9.10, 31.53] | 0.09 | 92.5 |
| Amikacin | 7 | 19.37 [4.93, 39.66] | 0.33 | 96.4 | 8 | 11.36 [4.03, 21.49] | 0.14 | 95.7 | 2 | 13.01 [2.83, 28.76] | 0.07 | 92.8 |
| Levofloxacin | 3 | 41.67 [10.31, 77.38] | 0.42 | 98.2 | 3 | 24.00 [9.29, 42.64] | 0.10 | 84.2 | 2 | 20.92 [11.16, 32.53] | 0.10 | 24.0 |
| Colistin | 4 | 18.62 [11.26, 27.25] | 0.03 | 76.8 | 3 | 61.26 [16.79, 96.48] | 0.68 | 98.3 | | - | - | - |
| Nalidixic Acid | 8 | 65.25 [47.87, 80.81] | 0.24 | 94.9 | 16 | 45.77 [31.70. 60.18] | 0.32 | 97.5 | 2 | 67.90 [32.43, 94.58] | 0.25 | 95.1 |
| Doxycycline | 5 | 49.58 [31.61, 67.59] | 0.14 | 90.7 | 4 | 60.06 [36.17, 81.79] | 0.20 | 91.2 | 2 | 28.09 [0.00, 99.68] | 1.41 | 95.6 |
| Lincomycin | | - | - | - | - | - | - | - | 2 | 22.34 [14.89, 30.72] | 0.00 | 0.0 |
| Neomycin | - | - | - | - | 3 | 20.35 [7.33, 37.28] | 0.08 | 73.88 | 3 | 13.06 [1.15, 33.76] | 0.18 | 93.6 |
| Penicillin | 3 | 75.64 [13.84, 100.0] | 1.30 | 98.2 | - | - | - | - | - | - | - | - |
| Kanamycin | 2 | 46.15 [29.35, 63.41] | 0.05 | 76.9 | 7 | 23.08 [15.29, 31.89] | 0.06 | 86.8 | 4 | 44.54 [16.19, 74.91] | 0.38 | 97.8 |
| Cefuroxime | 3 | 53.57 [10.86, 93.28] | 0.69 | 98.1 | 3 | 8.11 [0.06, 24.40] | 0.13 | 87.6 | - | - | - | - |
| Vancomycin | - | - | - | - | 2 | 63.89 [0.00, 100.00] | 2.41 | 97.4 | 2 | 7.18 [0.00, 25.22] | 0.11 | 89.5 |
| Cefoxitin | 4 | 19.00 [4.65, 39.45] | 0.19 | 93.0 | 3 | 5.03 [1.46, 10.39] | 0.02 | 81.4 | 4 | 31.59 [6.04, 65.31] | 0.47 | 98.2 |
| Oxacillin | - | - | - | - | - | - | - | - | 7 | 74.83 [57.74, 88.90] | 0.19 | 88.8 |
| Oxytetracycline | 4 | 72.77 [32.89, 98.82] | 0.62 | 96.3 | 5 | 71.28 [29.52, 99.17] | 0.80 | 95.1 | 2 | 55.76 [14.58, 92.68] | 0.40 | 96.8 |
| **Antibiotics** | ***Klebsiella* spp.** | | | | ***Enterococcus* spp.** | | | | ***Campylobacter* spp.** | | | |
| | **N** | **% (95 % CI)** | $\tau^2$ | $I^2$ | **N** | **% (95 % CI)** | $\tau^2$ | $I^2$ | **N** | **% (95 % CI)** | $\tau^2$ | $I^2$ |
| Amoxicillin | 2 | 90.91 [47.75, 100.0] | 0.48 | 95.9 | - | - | - | - | - | - | - | - |
| Ampicillin | 2 | 97.69 [61.69, 100.0] | 0.30 | 74.7 | - | - | - | - | 3 | 43.59 [18.76, 70.24] | 0.21 | 95.0 |
| Azithromycin | - | - | - | - | - | - | - | - | 2 | 8.20 [4.03, 13.41] | 0.00 | 0.00 |
| Chloramphenicol | - | - | - | - | 2 | 4.00 [0.00, 15.81] | 0.06 | 72.2 | 3 | 11.31 [0.00, 33.33] | 0.13 | 69.2 |
| Ciprofloxacin | - | - | - | - | 2 | 42.31 [33.14, 51.75] | 0.00 | 0.00 | 6 | 75.46 [58.99, 88.95] | 0.16 | 93.5 |
| Erythromycin | - | - | - | - | 3 | 37.32 [3.85, 80.10] | 0.59 | 97.2 | 4 | 6.01 [0.49, 15.35] | 0.06 | 83.1 |
| Streptomycin | - | - | - | - | 2 | 50.02 [40.15, 59.90] | 0.01 | 37.3 | 5 | 32.02 [3.29, 70.26] | 0.62 | 94.5 |
| Tetracycline | - | - | - | - | 2 | 51.35 [41.92, 60.73] | 0.00 | 0.00 | 6 | 50.40 [28.69, 72.04] | 0.24 | 94.0 |
| Gentamicin | 2 | 4.14 [0.00, 30.63] | 0.14 | 57.91 | - | - | - | - | 5 | 15.83 [6.16, 28.34] | 0.08 | 82.5 |
| Doxycycline | - | - | - | - | - | - | - | - | 2 | 21.49 [0.00, 69.65] | 0.38 | 82.3 |
| Nalidixic Acid | 2 | 30.23 [0.09, 75.89] | 0.30 | 74.6 | - | - | - | - | 7 | 65.63 [42.25, 85.83] | 0.34 | 96.2 |
| Neomycin | - | - | - | - | - | - | - | - | 3 | 17.28 [4.67, 34.94] | 0.10 | 82.4 |
| Tylosin | - | - | - | - | - | - | - | - | 2 | 79.72 [71.63, 86.92] | 0.00 | 0.0 |

* TMP-SMZ = Trimethoprim-Sulfamethoxazole, CI = Confidence Interval.

There are a few limitations in this study. First, incomplete reporting studies were not included, which may have led to bias in the findings, and the pooled prevalence estimates offer an overall summary, considerable heterogeneity among studies, including regarding sample types, geographic regions, and methodologies, necessitates cautious interpretation, as the results may not be generalizable to all populations. Second, although a comprehensive search and selection of eligible studies from Asia (2015–2025) was conducted in accordance with PRISMA guidelines, the pooled estimates may not fully represent the entire continent. Data were available from only 28 of approximately 47 Asian countries, with certain regions remaining underrepresented. Therefore, the findings should be interpreted with caution, as they primarily reflect the countries and subregions for which relevant data were accessible. Third, the systematic review protocol was not prospectively registered in PROSPERO or another registry. However, the review was conducted according to a predefined methodology following PRISMA guidelines, including systematic literature search, predefined inclusion and exclusion criteria, and a structured approach to data extraction and synthesis. We acknowledge that prospective registration could have increased transparency and confidence in the a priori design. Fourth, some AMR prevalence estimates in this study showed extremely wide confidence intervals, in certain cases spanning nearly the full range of 0–100 %. Such wide intervals indicate high uncertainty, likely due to small sample sizes, low incidence of resistance, or heterogeneity among the included studies. These unstable estimates should be interpreted with caution, and their presence highlights the need for larger, more representative datasets to generate more reliable prevalence data.

*Implications*

AMR is a critical global threat to human, animal, and environmental health, with the WHO listing it among the top 10 public health risks. If unchecked, AMR could cost the global economy up to $100 trillion. This study provides insights into the distribution and trends of bacteria isolated from meat and meat products, and AMR patterns with genotypes. These findings may inform public health strategies by indicating the region in Asia or the source of meat with the highest risk of AMR transmission. It also identified the more prevalent resistance gene responsible for AMR. Moreover, the important implications include the need for binding regulatory frameworks aligned with international standards, robust surveillance and reporting systems, incentives and support for producers, stringent stewardship guidelines, One Health coordination, and public and stakeholder engagement. Countries that have implemented such measures, such as the EU's growth-promoter ban and Denmark's yellow card scheme, have demonstrated substantial reductions in AMU and slowing AMR trends.

## Conclusion

Despite the guidelines that have been implemented worldwide, AMR

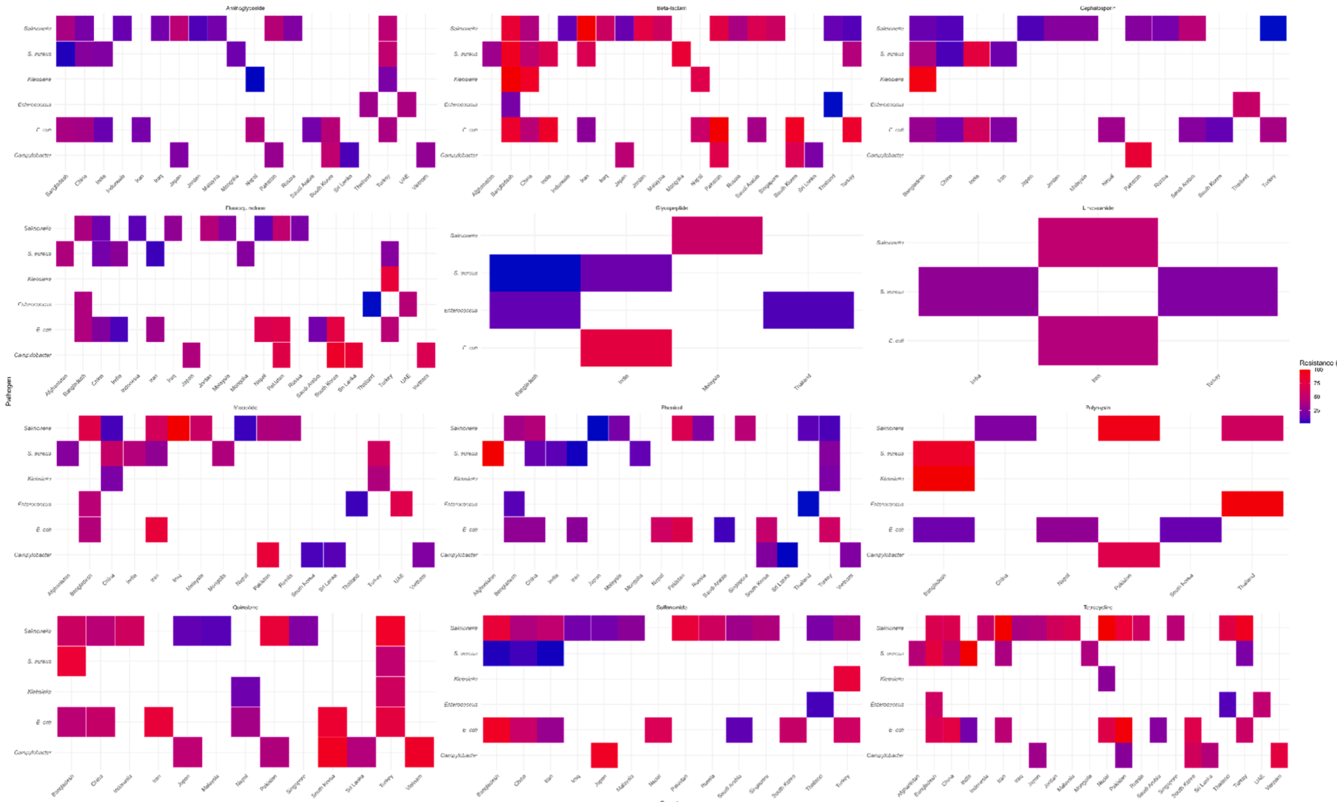

**Fig. 16.** Heatmap illustrating the distribution of antimicrobial resistance (AMR) patterns across bacterial species in various Asian countries. Each colored cell represents the prevalence of a specific resistance pattern within a bacterial species from a particular country. The color gradient ranges from blue (low prevalence) to red (high prevalence), with intermediate values shown in purple. This visualization allows rapid comparison of AMR patterns across countries and bacterial hosts.

is still a major recurrent global problem responsible for high morbidity rates. Antibiotics are used extensively in modern food animal production not only for therapeutic purposes but also to prevent disease and promote animal growth. Consequently, many healthy animals are frequently or regularly exposed to antibiotics. In summary, the findings support the necessity of strict rules and regulations to lower the use of antibiotics in meat across the whole food chain. Among the 28 countries, Myanmar presented the highest prevalence of bacteria isolated from meat, whereas *E. coli* was observed predominantly in Southeast Asia and South Asia. Therefore, these findings reveal increasing resistance of microorganisms to routinely administered antibiotics, greatly influencing the effectiveness of treatment. However, this scenario can be reduced by limiting antibiotic usage, continuing to update AMR trends, increasing public awareness, and following an efficient antibiotic regimen. In conclusion, we recommend more limited usage of antibiotics in the animal rearing sector and a more hygienic environment for meat vendors.

## Abbreviation

AMR Antimicrobial resistance
APAC Asia Pacific
MRSA Methicillin-resistant *Staphylococcus aureus*
MDR Multidrug Resistant
PRISMA Preferred Reporting Items for Systematic Reviews and Meta-Analyses
CI Confidence Interval
TMP-SMX Trimethoprim-Sulfamethoxazole
PCR Polymerase chain reaction
ARG Antibiotic resistance gene
CDC Centers for Disease Control and Prevention
AMU Antimicrobial use
*S. aureus* *Staphylococcus aureus*
*E. coli* *Escherichia coli*

## Clinical trial

Clinical trial number: not applicable

## Availability of data and materials

The datasets used and/or analyzed during the current study are available in the supplementary materials.

## Ethical approval and consent to participate

Not Applicable

## Consent for publication

Not Applicable

## Funding

There is no external or internal funding for this study.

## CRediT authorship contribution statement

**Md Jisan Ahmed:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Md Imran Hossain:** Writing – review & editing, Writing – original draft, Validation, Investigation, Data curation, Conceptualization. **Md Ismile Hossain Bhuiyan:** Writing –

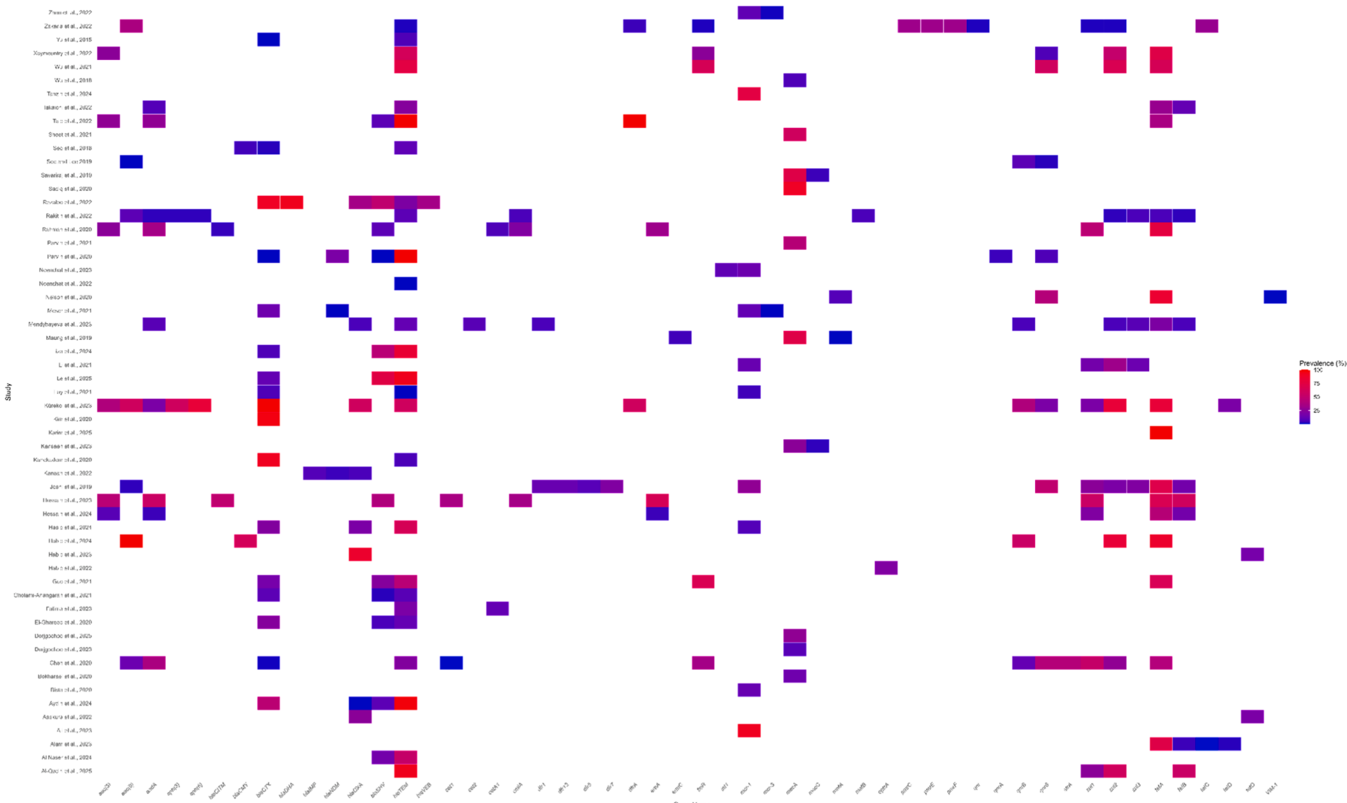


**Fig. 17.** Heatmap of antimicrobial resistance (AMR) genes reported across different studies. Each row represents a published study, and each column corresponds to a specific AMR gene. The color intensity indicates the reported prevalence of the gene, with red showing higher prevalence and blue showing lower prevalence. This visualization highlights the diversity and distribution of AMR genes across studies, with frequently reported genes appearing more consistently, while others show more sporadic occurrence.

review & editing, Writing – original draft, Validation, Data curation. **Md Arifur Rahman:** Writing – original draft, Validation, Data curation. **Israt Jahan Kaderi:** Validation, Data curation. **Ahmed Ishtiak Nirob:** Validation, Data curation. **Md Afiqul Islam Rahi:** Validation, Data curation. **Mustakim Mamun:** Validation, Data curation. **Kazi Estieque Alam:** Visualization, Software, Formal analysis. **Md Jayed Chowdhury:** Validation, Data curation. **Prajwal Bhandari:** Writing – review & editing, Writing – original draft. **Ritu Chalise:** Writing – review & editing, Writing – original draft. **Md Jahangir Alam:** Writing – review & editing, Validation. **Md Tazul Islam Chowdhury:** Writing – review & editing, Validation. **Mahabbat Ali:** Writing – review & editing, Validation. **Delower Hossain:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources.

## Disclosures

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

The author would like to acknowledge the technical support of the Association of Coding, Technology, and Genomics (ACTG), Sher-e-Bangla Agricultural University (SAU) members. The authors show their appreciation towards the University of Milan for providing literature support.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.psj.2025.105811.

## References

Abalaka, S., Sani, N., Idoko, I., Tenuche, O., Oyelowo, F., Ejeh, S., Enem, S., 2017. Pathological changes associated with an outbreak of colibacillosis in a commercial broiler flock. Sokoto J. Vet. Sci. 15 (3), 95–102.

Abd El Tawab, A.A., El-Hofy, F.I., Maarouf, A.A., El-Said, A.A., 2015. Bacteriological studies on some food borne bacteria isolated from chicken meat and meat products in Kaliobia Governorate. Benha Vet. Med. J. 29 (2), 47–59.

Abebaw, B., 2024. Prevalence of lumpy skin disease in Africa: a systematic review and meta-analysis from 2007 to 2023. Vet. Med. Int. 2024 (1), 9991106.

Abu-Madi, M., Behnke, J.M., Sharma, A., Bearden, R., Al-Banna, N., 2016. Prevalence of virulence/stress genes in Campylobacter jejuni from chicken meat sold in Qatari retail outlets. PLoS One 11 (6), e0156938. https://doi.org/10.1371/journal.pone.0156938.

Acharya, K.P., Wilson, R.T., 2019. Antimicrobial resistance in Nepal. Front. Med. 6, 105 (Lausanne).

Adiguzel, M.C., Baran, A., Wu, Z., Cengiz, S., Dai, L., Oz, C., Ozmenli, E., Goulart, D.B., Sahin, O., 2021. Prevalence of colistin resistance in *Escherichia coli* in Eastern Turkey and genomic characterization of an mcr-1 positive strain from retail chicken meat. Microb. Drug Resist. 27 (3), 424–432.

AHBD. (2025). Asia: How much do they consume? https://ahdb.org.uk/trade-and-policy/export-opps/regions/asia/consumption.

Ahmadi, S.A., Aziz, A., Afshar, M.F., Abi, A.J., 2025. Exploring staphylococcus aureus prevalence and antimicrobial resistance in ready-to-eat meat products in Kabul city. Egypt. J. Vet. Sci. 56 (7), 1–6.

Ahmed, M.O., Abouzeed, Y.M., Daw, M.A., 2025. Global initiatives to phase-out colistin use in food-producing animals. Open. Vet. J. 15 (2), 533–540. https://doi.org/10.5455/OVJ.2025.v15.i2.4.

Ahmed, S.K., Hussein, S., Qurbani, K., Ibrahim, R.H., Fareeq, A., Mahmood, K.A., Mohamed, M.G., 2024. Antimicrobial resistance: impacts, challenges, and future prospects. J. Med. Surg. Public Health 2, 100081.

Ain, Q.T., Arif, M., Ema, F.A., Shathi, M.S.A., Islam, M.A., Khatun, M.M., 2022. Multidrug-resistant Staphylococcus aureus isolated from chicken nuggets sold at superstores in Mymensingh, Bangladesh. J. Adv. Vet. Anim. Res. 9 (4), 601–609. https://doi.org/10.5455/javar.2022.i629.

Aklilu, E., Harun, A., Singh, K.K.B., 2022. Molecular characterization of bla.

Al-Hindi, R.R., Alharbi, M.G., Alotibi, I.A., Azhari, S.A., Ahmad, A., Alseghayer, M.S., Teklemariam, A.D., Almaneea, A.M., 2023. MALDI-TOF MS-based identification and

antibiotics profiling of Salmonella species isolated from retail chilled chicken in Saudi Arabia. J. King Saud Univ. - Sci. 35 (5), 102684. https://doi.org/10.1016/j.jksus.2023.102684.
Al-Qadiri, H.M., Al-Abdallat, A.M., Olaimat, A.N., Saleh, M., Bakri, F.G., Abutarbush, S.M., Rasco, B.A., 2025. Antimicrobial resistance of Salmonella spp. recovered from raw chicken meat at abattoirs and retail markets in Jordan. Ital. J. Food Sci./Riv. Ital. Sci. Alimenti 37 (2), 216–227.
Al Naser, J., Hossain, H., Chowdhury, M.S.R., Liza, N.A., Lasker, R.M., Rahman, A., Haque, M.A., Hossain, M.M., Rahman, M.M., 2024. Exploring of spectrum beta lactamase producing multidrug-resistant Salmonella enterica serovars in goat meat markets of Bangladesh. Vet. Anim. Sci. 25, 100367.
Alam, G.S., Hassan, M.M., Ahaduzzaman, M., Nath, C., Dutta, P., Khanom, H., Khan, S.A., Pasha, M.R., Islam, A., Magalhaes, R.S., 2023. Molecular detection of tetracycline-resistant genes in multi-drug-resistant *Escherichia coli* isolated from broiler meat in Bangladesh. Antibiotics 12 (2), 418.
Algammal, A.M., Hetta, H.F., Elkelish, A., Alkhalifah, D.H.H., Hozzein, W.N., Batiha, G.E.-S., El Nahhas, N., Mabrok, M.A., 2020. Methicillin-Resistant Staphylococcus aureus (MRSA): one health perspective approach to the bacterium epidemiology, virulence factors, antibiotic-resistance, and zoonotic impact. Infect. Drug Resist. 13. 3255–3265.
Alhabib, I., Elhadi, N., 2024. Antimicrobial resistance pattern of *Escherichia coli* isolated from imported frozen shrimp in Saudi Arabia. PeerJ. 12, e18689. https://doi.org/10.7717/peerj.18689.
Alhadlaq, M.A., Mujallad, M.I., Alajel, S.M.I., 2023. Detection of *Escherichia coli* O157:H7 in imported meat products from Saudi Arabian ports in 2017. Sci. Rep. 13 (1), 4222. https://doi.org/10.1038/s41598-023-30486-2.
Ali, F., Silvy, T. N., Hossain, T. J., Uddin, M. K., .and Uddin, M. S. (.2021). Prevalence and Antimicrobial Resistance Phenotypes of Salmonella Species Recovered at Various Stages of Broiler Operations in Hathazari, Bangladesh.
Ali, M.W., Utsho, K.S., Karmakar, S., Hoque, M.N., Rahman, M.T., Hassan, J., 2023. First report on the molecular characteristics of mcr-1 colistin resistant *E. coli* isolated from retail broiler meat in Bangladesh. Int. J. Food Microbiol. 388, 110065.
Ali, S., Alsayeqh, A.F., 2022. Review of major meat-borne zoonotic bacterial pathogens. Front. Public Health 10, 1045599.
Alkharusi, H., 2012. Categorical variables in regression analysis: a comparison of dummy and effect coding. Int. J. Educ. 4 (2), 202.
Altaf Hussain, M., Wang, W., Sun, C., Gu, L., Liu, Z., Yu, T., Ahmad, Y., Jiang, Z., Hou, J., 2020. Molecular characterization of pathogenic Salmonella spp from raw beef in Karachi, Pakistan. Antibiotics 9 (2), 73.
Amjad, M., Hassan, M.M., Raza, S., Nisar, M., Ahmad, M., Zia, U.-U.-R., 2023. Prevalence, risk factors analysis and antimicrobial resistance profile of Campylobacter species isolated from food animals in district Okara, Pakistan. Eur. J. Agric. Food Sci. 5 (1), 61–65.
Andrzejewska, M., Szczepańska, B., Śpica, D., Klawe, J.J., 2019. Prevalence, virulence, and antimicrobial resistance of *Campylobacter* spp. in raw milk, beef, and pork meat in Northern Poland. Foods 8 (9), 420.
Ansarifar, E., Riahi, S.M., Tasara, T., Sadighara, P., Zeinali, T., 2023. *Campylobacter* prevalence from food, animals, human and environmental samples in Iran: a systematic review and meta-analysis. BMC. Microbiol. 23 (1), 126.
Asakura, H., Yamamoto, S., Yamada, K., Kawase, J., Nakamura, H., Abe, K.-i., Sasaki, Y., Ikeda, T., Nomoto, R., 2022. Quantitative detection and genetic characterization of thermotolerant *Campylobacter* spp. in fresh chicken meats at retail in Japan [original research]. Front. Microbiol. 13, 2022. https://doi.org/10.3389/fmicb.2022.1014212.
Asif, M., Rahman, H., Qasim, M., Khan, T.A., Ullah, W., Jie, Y., 2017. Molecular detection and antimicrobial resistance profile of zoonotic Salmonella enteritidis isolated from broiler chickens in Kohat, Pakistan. J. Chin. Med. Assoc. 80 (5), 303–306.
Aslam, M., Nattress, F., Greer, G., Yost, C., Gill, C., McMullen, L., 2003. Origin of contamination and genetic diversity of *Escherichia coli* in beef cattle. Appl. Environ. Microbiol. 69 (5), 2794–2799. https://doi.org/10.1128/aem.69.5.2794-2799.2003.
Aydin, A., Suleymanoglu, A.A., Abdramanov, A., Paulsen, P., Dumen, E., 2024. Detection of extended spectrum ß-lactamase-producing *Escherichia coli* with biofilm formation from chicken meat in Istanbul. Foods 13 (7), 1122.
Azam, M., Mohsin, M., Saleemi, M.K., 2019. Virulence-associated genes and antimicrobial resistance among avian pathogenic *Escherichia coli* from colibacillosis affected broilers in Pakistan. Trop. Anim. Health Prod. 51, 1259–1265.
Baran, A., Adigüzel, M.C., Yüksel, M., 2020. Prevalence of antibiotic-resistant and extended-spectrum beta-lactamase-producing *Escherichia coli* in chicken meat from eastern Turkey. Pak. Vet. J. 40 (3), 2074–7764.
Baz, A.A., Bakhiet, E.K., Abdul-Raouf, U., Abdelkhalek, A., 2021. Prevalence of enterotoxin genes (SEA to SEE) and antibacterial resistant pattern of Staphylococcus aureus isolated from clinical specimens in Assiut city of Egypt. Egypt. J. Med. Hum. Genet. 22, 1–12.
Belotindos, L., Villanueva, M., Miguel, J., Bwalya, P., Harada, T., Kawahara, R., Nakajima, C., Mingala, C., Suzuki, Y., 2021. Prevalence and characterization of quinolone-resistance determinants in *Escherichia coli* isolated from food-producing animals and animal-derived food in the Philippines. Antibiotics 10 (4), 413. https://doi.org/10.3390/antibiotics10040413.
Bista, S., Thapa Shrestha, U., Dhungel, B., Koirala, P., Gompo, T.R., Shrestha, N., Adhikari, N., Joshi, D.R., Banjara, M.R., Adhikari, B., 2020. Detection of plasmid-mediated colistin resistant mcr-1 gene in *Escherichia coli* isolated from infected chicken livers in Nepal. Animals 10 (11), 2060.
Bivand, R., Altman, M., Anselin, L., Assunção, R., Berke, O., Bernat, A., and Blanchet, G. (2017). Package 'spdep'. Spatial dependence: Weighting schemes, statistics, R package version, 1.1–5.
Bokharaei, N.M., Dallal, M.S., Pourmand, M., Rajabi, Z., 2020. Antibiotic resistance pattern and detection of mecA gene in Staphylococcus aureus isolated from Iranian Hamburger samples. J. Food Qual. Hazards Control. 7 (4), 188–195.
Bottery, M.J., 2022. Ecological dynamics of plasmid transfer and persistence in microbial communities. Curr. Opin. Microbiol. 68, 102152.
Brown, A., Grass, J., Richardson, L., Nisler, A., Bicknese, A., Gould, L., 2017. Antimicrobial resistance in Salmonella that caused foodborne disease outbreaks: united States, 2003–2012. Epidemiol. Infect. 145 (4), 766–774.
Cao, H., Bougouffa, S., Park, T.-J., Lau, A., Tong, M.-K., Chow, K.-H., Ho, P.-L., 2022. Sharing of antimicrobial resistance genes between humans and food animals. mSystems 7 (6), e00775. -00722.
Chagas, V. d. S., Santos, C. d., Reis, W., Santos, A. d., Bezerra, M., and Seixas, V. (2017). Research Salmonella spp. in meat products from slaughterhouses Pará state in the 2014-2015 period.
Che, Y., Yang, Y., Xu, X., Břinda, K., Polz, M.F., Hanage, W.P., Zhang, T., 2021. Conjugative plasmids interact with insertion sequences to shape the horizontal transfer of antimicrobial resistance genes. Proc. Natl. Acad. Sci. U. S. A. 118 (6). https://doi.org/10.1073/pnas.2008731118.
Chen, Z., Bai, J., Wang, S., Zhang, X., Zhan, Z., Shen, H., Zhang, H., Wen, J., Gao, Y., Liao, M., Zhang, J., 2020. Prevalence, antimicrobial resistance, virulence genes and genetic diversity of Salmonella isolated from retail duck meat in southern China. Microorganisms 8 (3), 444. https://doi.org/10.3390/microorganisms8030444.
Conceição, S., Queiroga, M.C., Laranjo, M., 2023. Antimicrobial resistance in bacteria from meat and meat products: a one health perspective. Microorganisms (10), 11. https://doi.org/10.3390/microorganisms11102581.
Davis, G.S., Waits, K., Nordstrom, L., Weaver, B., Aziz, M., Gauld, L., Grande, H., Bigler, R., Horwinski, J., Porter, S., Stegger, M., Johnson, J.R., Liu, C.M., Price, L.B., 2015. Intermingled Klebsiella pneumoniae populations between retail meats and human urinary tract infections. Clin. Infect. Dis. 61 (6), 892–899. https://doi.org/10.1093/cid/civ428.
Dawadi, P., Bista, S., Bista, S., 2021. Prevalence of colistin-resistant *Escherichia coli* from poultry in South Asian developing countries. Vet. Med. Int. 2021 (1), 6398838.
Denton, M., O'Connell, B., Bernard, P., Jarlier, V., Williams, Z., Henriksen, A.S., 2008. The EPISA study: antimicrobial susceptibility of Staphylococcus aureus causing primary or secondary skin and soft tissue infections in the community in France, the UK and Ireland. J. Antimicrob. Chemother. 61 (3), 586–588.
DerSimonian, R., Laird, N., 1986. Meta-analysis in clinical trials. Control. Clin. Trials 7 (3), 177–188.
Devkota, S.P., Paudel, A., Gurung, K., 2019. Vancomycin intermediate MRSA isolates obtained from retail chicken meat and eggs collected at Pokhara, Nepal. Nepal J. Biotechnol. 7 (1), 90–95.
Dhaouadi, S., Soufi, L., Hamza, A., Fedida, D., Zied, C., Awadhi, E., Mtibaa, M., Hassen, B., Cherif, A., Torres, C., Abbassi, M.S., Landolsi, R.B., 2020. Co-occurrence of mcr-1 mediated colistin resistance and β-lactamase-encoding genes in multidrug-resistant *Escherichia coli* from broiler chickens with colibacillosis in Tunisia. J. Glob. Antimicrob. Resist. 22, 538–545. https://doi.org/10.1016/j.jgar.2020.03.017.
Dorjgochoo, A., Batbayar, A., Tsend-Ayush, A., Byambadorj, B., Jav, S., Yandag, M., 2025. Identification of Staphylococcus aureus causing contamination in raw beef and meat-processing environments in Ulaanbaatar, Mongolia. Int. J. Microbiol. 2025 (1), 3806846.
Dorjgochoo, A., Batbayar, A., Tsend-Ayush, A., Erdenebayar, O., Byambadorj, B., Jav, S., Yandag, M., 2023. Detection of virulence genes of Staphylococcus aureus isolated from raw beef for retail sale in the markets of Ulaanbaatar city, Mongolia. BMC. Microbiol. 23 (1), 372.
Doyle, C., Wall, K., Fanning, S., McMahon, B.J., 2025. Making sense of sentinels: wildlife as the one health bridge for environmental antimicrobial resistance surveillance. J. Appl. Microbiol. 136 (1), lxaf017. https://doi.org/10.1093/jambio/lxaf017.
El-Ghareeb, W.R., Abdel-Raheem, S.M., Al-Marri, T.M., Alaql, F.A., Fayez, M.M., 2020. Isolation and identification of extended spectrum β-lactamases (ESBLs) *Escherichia coli*from minced camel meat in Eastern province, Saudi Arabia. Thai J. Vet. Med. 50 (2), 155–161.
Eltai, N.O., Yassine, H.M., El-Obeid, T., Al-Hadidi, S.H., Al Thani, A.A., Alali, W.Q., 2020. Prevalence of antibiotic-resistant *Escherichia coli* isolates from local and imported retail chicken carcasses. J. Food Prot. 83 (12), 2200–2208. https://doi.org/10.4315/JFP-20-113.
Essack, S., 2021. Water, sanitation and hygiene in national action plans for antimicrobial resistance. Bull. World Health Organ. 99 (8), 606.
Fatima, A., Saleem, M., Nawaz, S., Khalid, L., Riaz, S., Sajid, I., 2023. Prevalence and antibiotics resistance status of Salmonella in raw meat consumed in various areas of Lahore, Pakistan. Sci. Rep. 13 (1), 22205. https://doi.org/10.1038/s41598-023-49487-2.
Feng, J., Wu, H., Zhuang, Y., Luo, J., Chen, Y., Wu, Y., Fei, J., Shen, Q., Yuan, Z., Chen, M., 2023. Stability and genetic insights of the co-existence of blaCTX-M-65, blaOXA-1, and mcr-1.1 harboring conjugative IncI2 plasmid isolated from a clinical extensively-drug resistant *Escherichia coli* ST744 in Shanghai [Original Research]. Front. Public Health 11, 2023. https://doi.org/10.3389/fpubh.2023.1216704.
Founou, L.L., Founou, R.C., Essack, S.Y., 2016. Antibiotic resistance in the food chain: a developing country-perspective. Front. Microbiol. 7, 1881.
Fowler, P.D., Sharma, S., Pant, D.K., Singh, S., Wilkins, M.J., 2021. Antimicrobial-resistant non-typhoidal Salmonella enterica prevalence among poultry farms and slaughterhouses in Chitwan, Nepal. Vet. World 14 (2), 437.
Furukawa, I., Ishihara, T., Teranishi, H., Saito, S., Yatsuyanagi, J., Wada, E., Kumagai, Y., Takahashi, S., Konno, T., Kashio, H., Kobayashi, A., Kato, N., Hayashi, K.-i., Fukushima, K., Ishikawa, K., Horikawa, K., Oishi, A., Izumiya, H., Ohnishi, T., Konishi, Y., Kuroki, T., 2017. Prevalence and characteristics of *Salmonella*and

campylobacterin retail poultry meat in Japan. Jpn. J. Infect. Dis. 70 (3), 239–247. https://doi.org/10.7883/yoken.JJID.2016.164.
Gholami-Ahangaran, M., Moravvej, A., Safizadeh, Z., Nogoorani, V.S., Zokaei, M., Ghasemian, S., 2021. The evaluation of ESBL genes and antibiotic resistance rate in *Escherichia coli* strains isolated from meat and intestinal contents of turkey in Isfahan, Iran. Iran. J. Vet. Res. 22 (4), 318.
Gonçalves-Tenório, A., Silva, B.N., Rodrigues, V., Cadavez, V., Gonzales-Barron, U., 2018. Prevalence of pathogens in poultry meat: a meta-analysis of European published surveys. Foods 7 (5), 69.
Grammatopoulos, T., Hunter, J.W., Munn, Z., Stone, J.C., Barker, T.H., 2023. Reporting quality and risk of bias in JBI systematic reviews evaluating the effectiveness of interventions: a methodological review protocol. JBI. Evid. Synth. 21 (3), 584–591.
Guo, S., Aung, K.T., Leekitcharoenphon, P., Tay, M.Y.F., Seow, K.L.G., Zhong, Y., Ng, L. C., Aarestrup, F.M., Schlundt, J., 2020. Prevalence and genomic analysis of ESBL-producing *Escherichia coli* in retail raw meats in Singapore. J. Antimicrob. Chemother. 76 (3), 601–605. https://doi.org/10.1093/jac/dkaa461.
Habib, I., Abdalla, A., Mohamed, M.-Y.I., Ghazawi, A., Khan, M., Elbediwi, M., Anes, F., Lakshmi, G.B., 2024. Genomic insights into antimicrobial resistant Salmonella in internationally traded chicken meat: first baseline findings in the United Arab Emirates. J. Agric. Food Res. 17, 101237. https://doi.org/10.1016/j.jafr.2024.101237.
Habib, I., Ibrahim Mohamed, M.-Y., Ghazawi, A., Lakshmi, G.B., Khan, M., Li, D., Sahibzada, S., 2023. Genomic characterization of molecular markers associated with antimicrobial resistance and virulence of the prevalent Campylobacter coli isolated from retail chicken meat in the United Arab Emirates. Curr. Res. Food Sci. 6, 100434. https://doi.org/10.1016/j.crfs.2023.100434.
Habib, I., Mohamed, M.-Y.I., Lakshmi, G.B., Khan, M., Li, D., 2022. Quantification of Campylobacter contamination on chicken carcasses sold in retail markets in the United Arab Emirates. Int. J. Food Contam. 9 (1), 9. https://doi.org/10.1186/s40550-022-00095-4.
Harrer, M., Cuijpers, P., Furukawa, T., Ebert, D., 2021. Doing Meta-analysis with R: a Hands-on Guide. Chapman and Hall/CRC.
Harun, A.B., Khatri, B., Karim, M.R., 2024. Phenotypic and genotypic pattern of antimicrobial resistance in livestock and poultry in South Asia: a systematic review and meta-analysis. Food Control. 164., 110575
Hasib, F.M.Y., Magouras, I., St-Hilaire, S., Paudel, S., Kamali, M., Lugsomya, K., Lam, H. K., Elsohaby, I., Butaye, P., Nekouei, O., 2024. Prevalence and characterization of antimicrobial-resistant *Escherichia coli* in chicken meat from wet markets in Hong Kong [Original Research]. Front. Vet. Sci. 11, 2024. https://doi.org/10.3389/fvets.2024.1340548.
Hayashi, W., Ohsaki, Y., Taniguchi, Y., Koide, S., Kawamura, K., Suzuki, M., Kimura, K., Wachino, J.-i., Nagano, Y., Arakawa, Y., Nagano, N., 2018. High prevalence of blaCTX-M-14 among genetically diverse *Escherichia coli* recovered from retail raw chicken meat portions in Japan. Int. J. Food Microbiol. 284, 98–104. https://doi.org/10.1016/j.ijfoodmicro.2018.08.003.
Hazards, E.P.o.B., Koutsoumanis, K., Allende, A., Álvarez-Ordóñez, A., Bolton, D., Bover-Cid, S., Chemaly, M., Davies, R., De Cesare, A., Herman, L., 2021. Role played by the environment in the emergence and spread of antimicrobial resistance (AMR) through the food chain. EFSA J. 19 (6), e06651.
Higgins, J.P., Thompson, S.G., 2002. Quantifying heterogeneity in a meta-analysis. Stat. Med. 21 (11), 1539–1558.
Hossain, M.M.K., Islam, M.S., Uddin, M.S., Rahman, A.M., Ud-Daula, A., Islam, M.A., Rubaya, R., Bhuiya, A.A., Alim, M.A., Jahan, N., 2022. Isolation, identification and genetic characterization of antibiotic resistant *Escherichia coli* from frozen chicken meat obtained from supermarkets at Dhaka City in Bangladesh. Antibiotics 12 (1), 41.
Hossain, M.S., Jony, M.A.H., Saha, N., Islam, B., Sobur, K.A., Mowdood, S., Hossain, K., 2024. Antibiotic resistance genes detection in *Escherichia coli* isolated from raw meat in Rajshahi division of Bangladesh. Am. J. Microbiol. Res. 12 (4), 79–84.
Huoy, L., Vuth, S., Hoeng, S., Chheang, C., Yi, P., San, C., Chhim, P., Thorn, S., Ouch, B., Put, D., 2024. Prevalence of Salmonella spp. in meat, seafood, and leafy green vegetables from local markets and vegetable farms in Phnom Penh, Cambodia. Food Microbiol. 124, 104614.
Hussain, M.A., Wang, W., Sun, C., Gu, L., Liu, Z., Yu, T., Ahmad, Y., Jiang, Z., Hou, J., 2020. Molecular characterization of pathogenic Salmonella spp from raw beef in Karachi, Pakistan. Antibiotics 9 (2), 73.
Ievy, S., Islam, M.S., Sobur, M.A., Talukder, M., Rahman, M.B., Khan, M.F.R., Rahman, M.T., 2020. Molecular detection of avian pathogenic *Escherichia coli* (APEC) for the first time in layer farms in Bangladesh and their antibiotic resistance patterns. Microorganisms 8 (7), 1021.
Islam, M., Sabrin, M.S., Kabir, M.H.B., Karim, S.J.I., Sikder, T., 2018. Prevalence of multidrug resistant (MDR) food-borne pathogens in raw chicken meat in Dhaka city, Bangladesh: an increasing food safety concern. Asian-Australas. J. Biosci. Biotechnol. 3 (1), 17–27.
Islam, M.S., Hasib, F.M.Y., Nath, C., Ara, J., Logno, T.A., Uddin, M.H., Khalil, M.I., Dutta, P., Das, T., Chowdhury, S., 2022. Molecular detection and risk factors associated with multidrug-resistant Campylobacter jejuni from broiler cloacal and meat samples in Bangladesh. Zoonoses. Public Health 69 (7), 843–855.
Islam, M.S., Hossain, M.J., Sobur, M.A., Punom, S.A., Rahman, A.T., Rahman, M.T., 2023. A systematic review on the occurrence of antimicrobial-resistant *Escherichia coli* in poultry and poultry environments in Bangladesh between 2010 and 2021. Biomed. Res. Int. 2023 (1), 2425564.
Islam, M.S., Nayeem, M.M.H., Sobur, M.A., Ievy, S., Islam, M.A., Rahman, S., Kafi, M.A., Ashour, H.M., Rahman, M.T., 2021. Virulence determinants and multidrug resistance of *Escherichia coli* isolated from migratory birds. Antibiotics 10 (2), 190.
Je, H.J., Singh, S., Kim, D.W., Hur, H.S., Kim, A.L., Seo, E.J., Koo, O.K., 2023. Systematic review and meta-analysis of Campylobacter species contamination in poultry, meat, and processing environments in South Korea. Microorganisms (11), 11. https://doi.org/10.3390/microorganisms11112722.
Joshi, P.R., Thummeepak, R., Leungtongkam, U., Pooarlai, R., Paudel, S., Acharya, M., Dhital, S., Sitthisak, S., 2019. The emergence of colistin-resistant *Escherichia coli* in chicken meats in Nepal. FEMS Microbiol. Lett. 366 (20), fnz237.
Julqarnain, S.M., Bose, P., Rahman, M.Z., Khatun, M.M., Islam, M.A., 2022. Bacteriological quality and prevalence of foodborne bacteria in broiler meat sold at live bird markets at Mymensingh City in Bangladesh. J. Adv. Vet. Anim. Res. 9 (3), 405–411. https://doi.org/10.5455/javar.2022.i608.
Kalpana, P., Yasobant, S., Saxena, D., Schreiber, C., 2024. Microbial contamination and antibiotic resistance in fresh produce and agro-ecosystems in South Asia—a systematic review. Microorganisms 12 (11), 2267.
Kanaan, M.H.G., Khalil, Z.K., Khashan, H.T., Ghasemian, A., 2022. Occurrence of virulence factors and carbapenemase genes in Salmonella enterica serovar Enteritidis isolated from chicken meat and egg samples in Iraq. BMC. Microbiol. 22 (1), 279.
Kang, C.-I., Song, J.-H., 2013. Antimicrobial resistance in Asia: current epidemiology and clinical implications. Infect. Chemother. 45 (1), 22.
Kang, H., Kim, H., Lee, J., Jeon, J.H., Kim, S., Park, Y., Joo, I., Kim, H., 2024. Genetic characteristics of multidrug-resistant Salmonella isolated from poultry meat in South Korea. Microorganisms 12 (8), 1646.
Kanokudom, S., Assawakongkarat, T., Akeda, Y., Ratthawongjirakul, P., Chuanchuen, R., Chaichanawongsaroj, N., 2021. Rapid detection of extended spectrum β-lactamase producing *Escherichia coli* isolated from fresh pork meat and pig cecum samples using multiplex recombinase polymerase amplification and lateral flow strip analysis. PLoS. One 16 (3), e0248536.
Kansaen, R., Boueroy, P., Hatrongjit, R., Kamjumphol, W., Kerdsin, A., Chopjitt, P., 2023. The occurrence and characteristics of methicillin-resistant staphylococcal isolates from foods and containers. Antibiotics 12 (8), 1287.
Karim, M.R., Zakaria, Z., Hassan, L., Faiz, N.M., Ahmad, N.I., 2023. The occurrence and molecular detection of mcr-1 and mcr-5 genes in Enterobacteriaceae isolated from poultry and poultry meats in Malaysia. Front. Microbiol. 14, 1208314.
Karim, S.J.I., Islam, K.S., Adnan, M.R., Sadi, M.A.H., Islam, M., 2025. Antibiotic-resistant Salmonella enteritidis in raw chicken meat of Dhaka City, Bangladesh. Int. J. Microbiol. 2025 (1), 5654730.
Kaushik, P., Anjay, A., Kumari, S., Dayal, S., Kumar, S., 2018. Antimicrobial resistance and molecular characterisation of *E. coli* from poultry in Eastern India. Vet. Ital. 54 (3), 197–204.
Khan, J.A., Rathore, R.S., Ahmad, I., Gill, R., Husain, F.M., Akhtar, J., 2022. High prevalence of multidrug resistant Staphylococcus aureus from Buffalo beef sold at retail butcheries in Northern India. Acta Sci. Microbiol. 5 (10), 12-21, (ISSN: 2581-3226).
Khan, S.A., Imtiaz, M.A., Sayeed, M.A., Shaikat, A.H., Hassan, M.M., 2020. Antimicrobial resistance pattern in domestic animal-wildlife-environmental niche via the food chain to humans with a Bangladesh perspective; a systematic review. BMC. Vet. Res. 16, 1–13.
Khan, S.B., Khan, M.A., Ahmad, I., ur Rehman, T., Ullah, S., Dad, R., Sultan, A., Memon, A.M., 2019. Phentotypic, gentotypic antimicrobial resistance and pathogenicity of Salmonella enterica serovars Typimurium and Enteriditis in poultry and poultry products. Microb. Pathog. 129, 118–124.
Khanal, S., Boonyayatra, S., Awaiwanont, N., 2022. Prevalence of methicillin-resistant Staphylococcus aureus in dairy farms: a systematic review and meta-analysis. Front. Vet. Sci. 9, 947154.
Kim, J., Park, H., Kim, J., Kim, J.H., Jung, J.I., Cho, S., Ryu, S., Jeon, B., 2019. Comparative analysis of aerotolerance, antibiotic resistance, and virulence gene prevalence in Campylobacter jejuni isolates from retail raw chicken and duck meat in South Korea. Microorganisms 7 (10), 433.
Kim, S., Kim, H., Kim, Y., Kim, M., Kwak, H., Ryu, S., 2020. Whole-genome sequencing-based characteristics in extended-spectrum beta-lactamase-producing *Escherichia coli* isolated from retail meats in Korea. Microorganisms 8 (4), 508.
Kirk, M.D., Pires, S.M., Black, R.E., Caipo, M., Crump, J.A., Devleesschauwer, B., Döpfer, D., Fazil, A., Fischer-Walker, C.L., Hald, T., 2015. World Health Organization estimates of the global and regional disease burden of 22 foodborne bacterial, protozoal, and viral diseases, 2010: a data synthesis. PLoS. Med. 12 (12), e1001921.
Koczura, R., Mokracka, J., Taraszewska, A., Łopacinska, N., 2016. Abundance of class 1 integron-integrase and sulfonamide resistance genes in river water and sediment is affected by anthropogenic pressure and environmental factors. Microb. Ecol. 72 (4), 909–916. https://doi.org/10.1007/s00248-016-0843-4.
Kottawatta, K.S., Van Bergen, M.A., Abeynayake, P., Wagenaar, J.A., Veldman, K.T., Kalupahana, R.S., 2017. Campylobacter in broiler chicken and broiler meat in Sri Lanka: influence of semi-automated vs. wet market processing on Campylobacter contamination of broiler neck skin samples. Foods 6 (12), 105.
Kumar, S., 2021. Antimicrobial resistance: a top ten global public health threat. EClinicalMedicine 41, 101221.
Kümmerer, K., 2009. Antibiotics in the aquatic environment–a review–part I. Chemosphere 75 (4), 417–434. https://doi.org/10.1016/j.chemosphere.2008.11.086.
Kürekci, C., Ünaldı, Ö., Şahin, S., García-Meniño, I., Hammerl, J.A., 2023. Impact and diversity of ESBL-producing Klebsiella pneumoniae recovered from raw chicken meat samples in Türkiye. Antibiotics 13 (1), 14.
Kusi, J., Ojewole, C.O., Ojewole, A.E., Nwi-Mozu, I., 2022. Antimicrobial resistance development pathways in surface waters and public health implications. Antibiotics 11 (6), 821.

Lay, K.K., Jeamsripong, S., Sunn, K.P., Angkititrakul, S., Prathan, R., Srisanga, S., Chuanchuen, R., 2021. Colistin resistance and ESBL production in Salmonella and *Escherichia coli* from pigs and pork in the Thailand, Cambodia, Lao PDR, and Myanmar border area. Antibiotics 10 (6), 657.

Le, Y.H., Hoang, H.T.T., Khong, D.T., Nguyen, T.N., Que, T.A., Pham, D.T., Tanaka, K., Yamamoto, Y., 2025. Contamination of retail market meat with extended-spectrum beta-lactamase genes in Vietnam. Int. J. Food Microbiol. 430, 111061. https://doi.org/10.1016/j.ijfoodmicro.2025.111061.

Li, H., Liu, Y., Yang, L., Wu, X., Wu, Y., Shao, B., 2021. Prevalence of *Escherichia coli* and antibiotic resistance in animal-derived food samples - six districts, Beijing, China, 2020. China CDC. Wkly. 3 (47), 999–1004. https://doi.org/10.46234/ccdcw2021.243.

Li, L., Heidemann Olsen, R., Ye, L., Yan, H., Nie, Q., Meng, H., Shi, L., 2016. Antimicrobial resistance and resistance genes in aerobic bacteria isolated from pork at slaughter. J. Food Prot. 79 (4), 589–597. https://doi.org/10.4315/0362-028x.Jfp-15-455.

Li, N., Liu, C., Zhang, Z., Li, H., Song, T., Liang, T., Li, B., Li, L., Feng, S., Su, Q., 2019. Research and technological advances regarding the study of the spread of antimicrobial resistance genes and antimicrobial-resistant bacteria related to animal husbandry. Int. J. Environ. Res. Public Health 16 (24), 4896.

Liaqat, Z., Khan, I., Azam, S., Anwar, Y., Althubaiti, E.H., Maroof, L., 2022. Isolation and molecular characterization of extended spectrum beta lactamase producing *Escherichia coli* from chicken meat in Pakistan. PLoS One 17 (6), e0269194.

Liu, G., Ali, T., Gao, J., Ur Rahman, S., Yu, D., Barkema, H.W., Huo, W., Xu, S., Shi, Y., Kastelic, J.P., Han, B., 2020. Co-occurrence of plasmid-mediated colistin resistance (mcr-1) and extended-spectrum β-lactamase encoding genes in *Escherichia coli* from bovine mastitic milk in China. Microb. Drug Resist. 26 (6), 685–696. https://doi.org/10.1089/mdr.2019.0333.

Liu, Y., Duan, H., Yang, L., Chen, H., Wu, R., Li, Y., Zhu, Y., Li, J., 2024. Antimicrobial resistance profiling of pathogens from cooked donkey meat products in Beijing area in one health context. Vet. Sci. 11 (12), 645. https://doi.org/10.3390/vetsci11120645.

Liza, N.A., Hossain, H., Rahman Chowdhury, M.S., Al Naser, J., Lasker, R.M., Rahman, A., Haque, M.A., Al Mamun, M., Hossain, M.M., Rahman, M.M., 2024. Molecular epidemiology and antimicrobial resistance of extended-spectrum β-lactamase (ESBL)-producing Klebsiella pneumoniae in retail cattle meat. Vet. Med. Int. 2024, 3952504. https://doi.org/10.1155/2024/3952504.

Mahato, S., 2019. Relationship of sanitation parameters with microbial diversity and load in raw meat from the outlets of the Metropolitan City Biratnagar, Nepal. Int. J. Microbiol. 2019 (1), 3547072.

Makwana, P., Nayak, J., Brahmbhatt, M., Chaudhary, J., 2015. Detection of Salmonella spp. From chevon, mutton and its environment in retail meat shops in Anand city (Gujarat), India. Vet. World 8 (3), 388.

Martinez, J.L., 2009. Environmental pollution by antibiotics and by antibiotic resistance determinants. Environ. Pollut. 157 (11), 2893–2902. https://doi.org/10.1016/j.envpol.2009.05.051.

Matheou, A., Abousetta, A., Pascoe, A.P., Papakostopoulos, D., Charalambous, L., Panagi, S., Panagiotou, S., Yiallouris, A., Filippou, C., Johnson, E.O., 2025. Antibiotic use in livestock farming: a driver of multidrug resistance? Microorganisms 13 (4). https://doi.org/10.3390/microorganisms13040779.

Mendybayeva, A., Abilova, Z., Bulashev, A., Rychshanova, R., 2023. Prevalence and resistance to antibacterial agents in Salmonella enterica strains isolated from poultry products in Northern Kazakhstan. Vet. World 16 (3), 657.

Mir, R., Salari, S., Najimi, M., Rashki, A., 2022. Determination of frequency, multiple antibiotic resistance index and resistotype of Salmonella spp. in chicken meat collected from southeast of Iran. Vet. Med. Sci. 8 (1), 229–236.

Moe, A.Z., Paulsen, P., Pichpol, D., Fries, R., Irsigler, H., Baumann, M.P., Oo, K.N., 2017. Prevalence and antimicrobial resistance of Salmonella isolates from chicken carcasses in retail markets in Yangon, Myanmar. J. Food Prot. 80 (6), 947–951.

Moher, D., Liberati, A., Tetzlaff, J., Altman, D.G., 2009. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. 339 (7716). BMJ 2535.

Mondal, A.H., Khare, K., Saxena, P., Debnath, P., Mukhopadhyay, K., Yadav, D., 2024. A review on colistin resistance: an antibiotic of last resort. Microorganisms 12 (4), 772.

Monte, D.F., Fernandes, M.R., Cerdeira, L., Esposito, F., Galvão, J.A., Franco, B.D., Lincopan, N., Landgraf, M., 2017. Chicken meat as a reservoir of colistin-resistant *Escherichia coli* strains carrying mcr-1 genes in South America. Antimicrob. Agents Chemother. 61 (5). https://doi.org/10.1128/aac.02718-02716.

Mori, T., Okamura, N., Kishino, K., Wada, S., Zou, B., Nanba, T., Ito, T., 2018. Prevalence and antimicrobial resistance of salmonellaserotypes isolated from poultry meat in Japan. Food Saf. 6 (3), 126–129. https://doi.org/10.14252/foodsafetyfscj.2017019.

Moser, A.I., Kuenzli, E., Campos-Madueno, E.I., Büdel, T., Rattanavong, S., Vongsouvath, M., Hatz, C., Endimiani, A., 2021. Antimicrobial-resistant *Escherichia coli* strains and their plasmids in people, poultry, and chicken meat in Laos. Front. Microbiol. 12, 708182.

Muinde, P., Maina, J., Momanyi, K., Yamo, V., Mwaniki, J., Kiiru, J., 2023. Antimicrobial resistant pathogens detected in raw pork and poultry meat in retailing outlets in Kenya. Antibiotics 12 (3), 613. https://www.mdpi.com/2079-6382/12/3/613.

Mulchandani, R., Wang, Y., Gilbert, M., Van Boeckel, T.P., 2023. Global trends in antimicrobial use in food-producing animals: 2020 to 2030. PLOS Glob. Public Health 3 (2), e0001305. https://doi.org/10.1371/journal.pgph.0001305.

Murray, C.J., Ikuta, K.S., Sharara, F., Swetschinski, L., Aguilar, G.R., Gray, A., Han, C., Bisignano, C., Rao, P., Wool, E., 2022. Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. Lancet 399 (10325), 629–655. https://doi.org/10.1016/S0140-6736(21)02724-0.

Naas, T., Aubert, D., Lambert, T., Nordmann, P., 2006. Complex genetic structures with repeated elements, a sul-type class 1 integron, and the blaVEB extended-spectrum beta-lactamase gene. Antimicrob. Agents Chemother. 50 (5), 1745–1752. https://doi.org/10.1128/aac.50.5.1745-1752.2006.

Nadimpalli, M., Vuthy, Y., de Lauzanne, A., Fabre, L., Criscuolo, A., Gouali, M., Huynh, B.-T., Naas, T., Phe, T., Borand, L., 2019. Meat and fish as sources of extended-spectrum β-lactamase–producing *Escherichia coli*, Cambodia. Emerg. Infect. Dis. 25 (1), 126.

Nagpala, M.J.M., Mora, J.F.B., Pavon, R.D.N., Rivera, W.L., 2025. Genomic characterization of antimicrobial-resistant Salmonella enterica in chicken meat from wet markets in Metro Manila, Philippines [original research]. Front. Microbiol. 16, 2025. https://doi.org/10.3389/fmicb.2025.1496685.

Naik, V., Shakya, S., Patyal, A., Gade, N., 2015. Isolation and molecular characterization of Salmonella spp. from chevon and chicken meat collected from different districts of Chhattisgarh, India. Vet. World 8 (6), 702.

Nellums, L.B., Thompson, H., Holmes, A., Castro-Sánchez, E., Otter, J.A., Norredam, M., Friedland, J.S., Hargreaves, S., 2018. Antimicrobial resistance among migrants in Europe: a systematic review and meta-analysis. Lancet Infect. Dis. 18 (7), 796–811. https://doi.org/10.1016/s1473-3099(18)30219-6.

Nelson, A., Manandhar, S., Ruzante, J., Gywali, A., Dhakal, B., Dulal, S., Chaulagai, R., Dixit, S.M., 2020. Antimicrobial drug resistant non-typhoidal Salmonella enterica in commercial poultry value chain in Chitwan, Nepal. One Health Outlook 2, 1–8.

Neogi, S.B., Islam, M.M., Islam, S.S., Akhter, A.T., Sikder, M.M.H., Yamasaki, S., Kabir, S. L., 2020. Risk of multi-drug resistant *Campylobacter* spp. And residual antimicrobials at poultry farms and live bird markets in Bangladesh. BMC. Infect. Dis. 20, 1–14.

Neyra, R.C., Vegosen, L., Davis, M.F., Price, L., Silbergeld, E.K., 2012. Antimicrobial-resistant bacteria: an unrecognized work-related risk in food animal production. Saf. Health Work 3 (2), 85–91.

Nguyen, T.N.M., Hotzel, H., El-Adawy, H., Tran, H.T., Le, M.T.H., Tomaso, H., Neubauer, H., Hafez, H.M., 2016. Genotyping and antibiotic resistance of thermophilic Campylobacter isolated from chicken and pig meat in Vietnam. Gut. Pathog. 8 (1), 19. https://doi.org/10.1186/s13099-016-0100-x.

Nhung, N.T., Cuong, N.V., Thwaites, G., Carrique-Mas, J., 2016. Antimicrobial usage and antimicrobial resistance in animal production in Southeast Asia: a review. Antibiotics 5 (4), 37.

Nhung, N.T., Phu, D.H., Carrique-Mas, J.J., Padungtod, P., 2024. A review and meta-analysis of non-typhoidal Salmonella in Vietnam: challenges to the control and antimicrobial resistance traits of a neglected zoonotic pathogen. One Health. 18., 100698

Nisansala, T., Gunasekara, Y.D., Piyarathne, N.S., 2025. Phenotypic and genotypic landscape of antibiotic resistance through one Health approach in Sri Lanka: a systematic review. Trop. Med. Int. Health 30 (3), 143–158. https://doi.org/10.1111/tmi.14084.

Nisar, M., Mushtaq, M.H., Shehzad, W., Hussain, A., Muhammad, J., Nagaraja, K.V., Goyal, S.M., 2017. Prevalence and antimicrobial resistance patterns of *Campylobacter* spp. Isolated from retail meat in Lahore, Pakistan. Food Control 80, 327–332.

Noenchat, P., Direksin, K., Sornplang, P., 2023. The phenotypic and genotypic antimicrobial resistance patterns of Salmonella isolated from chickens and meat at poultry slaughterhouses in Japan and Thailand. Vet. World 16 (7), 1527.

Noenchat, P., Nhoonoi, C., Srithong, T., Lertpiriyasakulkit, S., Sornplang, P., 2022. Prevalence and multidrug resistance of Enterococcus species isolated from chickens at slaughterhouses in Nakhon Ratchasima Province, Thailand. Vet. World 15 (11), 2535.

Nwobodo, D.C., Ugwu, M.C., Oliseloke Anie, C., Al-Ouqaili, M.T., Chinedu Ikem, J., Victor Chigozie, U., Saki, M., 2022. Antibiotic resistance: the challenges and some emerging strategies for tackling a global menace. J. Clin. Lab. Anal. 36 (9), e24655.

Odetokun, I.A., Adetona, M.A., Ade-Yusuf, R.O., Adewoye, A.O., Ahmed, A.N., Ghali-Mohammed, I., Al-Mustapha, A.I., Fetsch, A., 2023. Staphylococcus aureus contamination of animal-derived foods in Nigeria: a systematic review, 2002—2022. Food Saf. Risk 10 (1), 6.

Odeyemi, O.A., Alegbeleye, O.O., Strateva, M., Stratev, D., 2020. Understanding spoilage microbial community and spoilage mechanisms in foods of animal origin. Compr. Rev. Food Sci. Food Saf. 19 (2), 311–331.

Olaitan, A.O., Dandachi, I., Baron, S.A., Daoud, Z., Morand, S., Rolain, J.-M., 2021. Banning colistin in feed additives: a small step in the right direction. Lancet Infect. Dis. 21 (1), 29–30.

Oswaldi, V., Lüth, S., Dzierzon, J., Meemken, D., Schwarz, S., Feßler, A.T., Félix, B., Langforth, S., 2022. Distribution and characteristics of Listeria spp. in pigs and pork production chains in Germany. Microorganisms 10 (3), 512.

Parvin, M.S., Ali, M.Y., Talukder, S., Nahar, A., Chowdhury, E.H., Rahman, M.T., Islam, M.T., 2021. Prevalence and multidrug resistance pattern of methicillin resistant S. aureus isolated from frozen chicken meat in Bangladesh. Microorganisms 9 (3). https://doi.org/10.3390/microorganisms9030636.

Parvin, M.S., Hasan, M.M., Ali, M.Y., Chowdhury, E.H., Rahman, M.T., Islam, M.T., 2020. Prevalence and multidrug resistance pattern of Salmonella carrying extended-spectrum β-lactamase in frozen chicken meat in Bangladesh. J. Food Prot. 83 (12), 2107–2121.

Parvin, M.S., Talukder, S., Ali, M.Y., Chowdhury, E.H., Rahman, M.T., Islam, M.T., 2020. Antimicrobial resistance pattern of *Escherichia coli* isolated from frozen chicken meat in Bangladesh. Pathogens 9 (6), 420.

Paudyal, N., Pan, H., Liao, X., Zhang, X., Li, X., Fang, W., Yue, M., 2018. A meta-analysis of major foodborne pathogens in Chinese food commodities between 2006 and 2016. Foodborne Pathog. Dis. 15 (4), 187–197.

Paul, D., Das, B., 2022. Gut microbiome in the emergence of antibiotic-resistant bacterial pathogens. Prog. Mol. Biol. Transl. Sci. 192 (1), 1–31.

Pehlivanlar Önen, S., Aslantaş, Ö., Şebnem Yılmaz, E., Kürekci, C., 2015. Prevalence of β-lactamase producing *Escherichia coli* from retail meat in Turkey. J. Food Sci. 80 (9), M2023–M2029.

Peng, M., Salaheen, S., Biswas, D., 2014. Animal health: global antibiotic issues. In N. K. Van Alfen (Ed.), Encyclopedia of Agriculture and Food Systems (pp. 346-357). Academic Press.

Pérez-Boto, D., D'Arrigo, M., García-Lafuente, A., Bravo, D., Pérez-Baltar, A., Gaya, P., Medina, M., Arqués, J.L., 2023. Staphylococcus aureus in the processing environment of cured meat products. Foods (11), 12. https://doi.org/10.3390/foods12112161.

Poey, M.E., Azpiroz, M.F., Laviña, M., 2019. On sulfonamide resistance, sul genes, class 1 integrons and their horizontal transfer in *Escherichia coli*. Microb. Pathog. 135, 103611. https://doi.org/10.1016/j.micpath.2019.103611.

Premarathne, J.M., Anuar, A.S., Thung, T.Y., Satharasinghe, D.A., Jambari, N.N., Abdul-Mutalib, N.-A., Huat, J.T.Y., Basri, D.F., Rukayadi, Y., Nakaguchi, Y., 2017. Prevalence and antibiotic resistance against tetracycline in Campylobacter jejuni and C. coli in cattle and beef meat from Selangor, Malaysia. Front. Microbiol. 8, 2254.

Priya, G.B., Srinivas, K., Shilla, H., Milton, A.A.P., 2023. High prevalence of multidrug-resistant, biofilm-forming virulent Clostridium perfringens in broiler chicken retail points in Northeast India. Foods 12 (22), 4185.

Rabby, M.R.I., Shah, S.T., Miah, M.I., Islam, M.S., Khan, M.A.S., Rahman, M.S., Malek, M. A., 2021. Comparative analysis of bacteriological hazards and prevalence of Salmonella in poultry-meat retailed in wet-and super-markets in Dhaka city, Bangladesh. J. Agric. Food Res. 6, 100224.

Radi, N.K., Al-Marzoqi, A.H., 2023. The distribution of antimicrobial resistance and the presence of virulence genes in *Escherichia coli* isolated from frozen chicken meat in Iraq. AIP. Conf. Proc.2776 1, 020003.

Rafiq, K., Sani, A.A., Hossain, M.T., Hossain, M.T., Hadiuzzaman, M., Bhuiyan, M.A.S., 2024. Assessment of the presence of multidrug-resistant *Escherichia coli*, Salmonella and Staphylococcus in chicken meat, eggs and faeces in Mymensingh division of Bangladesh. Heliyon 10 (17), e36690.

Rahman, M., Ahmed, M., 2022. Antibiogram of E. coli and Salmonella spp. Isolated from chicken meat and frozen milk in Barishal city, Bangladesh. Bangladesh J. Vet. Med. (BJVM) 20 (1), 49–58.

Rahman, M., Rahman, A., Islam, M., Alam, M., 2018. Detection of multi–drug resistant Salmonella from milk and meat in Bangladesh. Bangladesh J. Vet. Med. 16 (1), 115–120.

Rahman, M.M., Husna, A., Elshabrawy, H.A., Alam, J., Runa, N.Y., Badruzzaman, A., Banu, N.A., Al Mamun, M., Paul, B., Das, S., 2020. Isolation and molecular characterization of multidrug-resistant *Escherichia coli* from chicken meat. Sci. Rep. 10 (1), 21999.

Rajaei, M., Moosavy, M.-H., Gharajalar, S.N., Khatibi, S.A., 2021. Antibiotic resistance in the pathogenic foodborne bacteria isolated from raw kebab and hamburger: phenotypic and genotypic study. BMC. Microbiol. 21, 1–16.

Rakitin, A.L., Yushina, Y.K., Zaiko, E.V., Bataeva, D.S., Kuznetsova, O.A., Semenova, A. A., Ermolaeva, S.A., Beletskiy, A.V., Kolganova, T.y.V., Mardanov, A.V., Shapovalov, S.O., Tkachik, T.E., 2022. Evaluation of antibiotic resistance of Salmonella serotypes and whole-genome sequencing of multiresistant strains isolated from food products in Russia. Antibiotics 11 (1), 1. https://doi.org/10.3390/antibiotics11010001.

Ranasinghe, R.A.S.S., Satharasinghe, D.A., Anwarama, P.S., Parakatawella, P.M.S.D.K., Jayasooriya, L.J.P.A.P., Ranasinghe, R.M.S.B.K., Rajapakse, R.P.V.J., Huat, J.T.Y., Rukayadi, Y., Nakaguchi, Y., Nishibuchi, M., Radu, S., Premarathne, J.M.K.J.K., 2022. Prevalence and antimicrobial resistance of *Escherichia coli* in chicken meat and edible poultry organs collected from retail shops and supermarkets of North Western Province in Sri Lanka. J. Food Qual. 2022 (1), 8962698. https://doi.org/10.1155/2022/8962698.

Ren, X., Yang, D., Yang, Z., Li, Y., Yang, S., Li, W., Qiao, X., Xue, C., Chen, M., Zhang, L., Yan, L., Peng, Z., 2024. Prevalence and antimicrobial susceptibility of foodborne pathogens from raw livestock meat in China, 2021. Microorganisms 12 (11), 2157. https://doi.org/10.3390/microorganisms12112157.

Ripon, R.K., Motahara, U., Ahmed, A., Devnath, N., Mahua, F.A., Hashem, R.B., Ishadi, K. S., Alam, A., Sujan, M.S.H., Sarker, M.S., 2023. Exploring the prevalence of antibiotic resistance patterns and drivers of antibiotics resistance of Salmonella in livestock and poultry-derived foods: a systematic review and meta-analysis in Bangladesh from 2000 to 2022. JAC. Antimicrob. Resist. 5 (3), dlad059.

Rortana, C., Nguyen-Viet, H., Tum, S., Unger, F., Boqvist, S., Dang-Xuan, S., Koam, S., Grace, D., Osbjer, K., Heng, T., 2021. Prevalence of Salmonella spp. And Staphylococcus aureus in chicken meat and pork from Cambodian markets. Pathogens 10 (5), 556.

Sadiq, A., Samad, M., Saddam, B.a., Ali, N., Roohullah, S., Saad, Z., Khan, A.N., Ahmad, Y., Khan, A., 2020. Methicillin-resistant Staphylococcus aureus (MRSA) in slaughter houses and meat shops in capital territory of Pakistan during 2018–2019. Front. Microbiol. 11, 577707.

Samad, A., Abbas, F., Ahmad, Z., Tanveer, Z., Ahmad, I., Patching, S.G., Nawaz, N., Asmat, M.T., Raziq, A., Asadullah, 2018. Multiplex polymerase chain reaction detection of Shiga toxin genes and antibiotic sensitivity of *Escherichia coli* O157: H7 isolated from beef meat in Quetta, Pakistan. J. Food Saf. 38 (6), e12540.

Samad, M.A., Eberson, L., Begum, R., Alam, M.G.S., Talukdar, F., Akter, R., Dang-Xuan, S., Sharma, G., Islam, S., Siddiky, N.A., 2023. Microbial contamination and antibiotic resistance in marketed food in Bangladesh: current situation and possible improvements. Antibiotics 12 (3), 555.

Samad, M.A., Sagor, M.S., Hossain, M.S., Karim, M.R., Mahmud, M.A., Sarker, M.S., Shownaw, F.A., Mia, Z., Card, R.M., Agunos, A., 2022. High prevalence of vancomycin non-susceptible and multi-drug resistant enterococci in farmed animals and fresh retail meats in Bangladesh. Vet. Res. Commun. 46 (3), 811–822.

Şanlıbaba, P., 2022. Prevalence, antibiotic resistance, and enterotoxin production of Staphylococcus aureus isolated from retail raw beef, sheep, and lamb meat in Turkey. Int. J. Food Microbiol. 361, 109461.

Santos, P.D.M., Widmer, K.W., Rivera, W.L., 2020. PCR-based detection and serovar identification of Salmonella in retail meat collected from wet markets in Metro Manila, Philippines. PLoS One 15 (9), e0239457. https://doi.org/10.1371/journal.pone.0239457.

Saud, B., Paudel, G., Khichaju, S., Bajracharya, D., Dhungana, G., Awasthi, M.S., Shrestha, V., 2019. Multidrug-resistant bacteria from raw meat of buffalo and chicken, Nepal. Vet. Med. Int. 2019 (1), 7960268.

Savariraj, W.R., Ravindran, N.B., Kannan, P., Paramasivam, R., Senthilkumar, T., Kumarasamy, P., Rao, V.A., 2019. Prevalence, antimicrobial susceptibility and virulence genes of Staphylococcus aureus isolated from pork meat in retail outlets in India. J. Food Saf. 39 (1), e12589.

Seo, K.W., Kim, Y.B., Jeon, H.Y., Lim, S.-K., Lee, Y.J., 2018. Comparative genetic characterization of third-generation cephalosporin-resistant *Escherichia coli* from chicken meat produced by integrated broiler operations in South Korea. Poult. Sci. 97 (8), 2871–2879.

Seo, K.W., Lee, Y.J., 2019. Prevalence and characterization of plasmid mediated quinolone resistance genes and class 1 integrons among multidrug-resistant *Escherichia coli* isolates from chicken meat. J. Appl. Poult. Res. 28 (3), 761–770.

Shabbir, M.A.B., Ul-Rahman, A., Iftikhar, M.R., Rasheed, M., Maan, M.K., Sattar, A., Ahmad, M., Khan, F.A., Ahmad, W., Riaz, M.I., 2024. Exploring the interplay of the CRISPR-CAS system with antibiotic resistance in Staphylococcus aureus: a poultry meat study from lahore, Pakistan. Medicina 60 (1), 130 (B Aires).

Shakhes, S.A., Wasim, W.A., Nasiry, Z., Tookhy, N.A., 2023. Coliform contamination of raw beef at the slaughterhouse and butchery levels in herat city, Afghanistan. Nangarhar Univ. Int. J. Biosci. 2 (04), 137–144.

Sheet, O., Hussien, S., and Alchalaby, A. (2021). Detection of methicillin-resistant Staphylococcus aureus from broiler carcasses in mosul city.

Shrestha, A., Bajracharya, A.M., Subedi, H., Turha, R.S., Kafle, S., Sharma, S., Neupane, S., Chaudhary, D.K., 2017. Multi-drug resistance and extended spectrum beta lactamase producing gram negative bacteria from chicken meat in Bharatpur Metropolitan, Nepal. BMC. Res. Notes 10, 1–5.

Silva, V., Araújo, S., Monteiro, A., Eira, J., Pereira, J.E., Maltez, L., Igrejas, G., Lemsaddek, T.S., Poeta, P., 2023. Staphylococcus aureus and MRSA in livestock: antimicrobial resistance and genetic lineages. Microorganisms 11 (1), 124.

Siriken, B., Türk, H., Yildirim, T., Durupinar, B., Erol, I., 2015. Prevalence and characterization of Salmonella isolated from chicken meat in Turkey. J. Food Sci. 80 (5), M1044–M1050.

Soepranianondo, K., Wardhana, D.K., 2019. Analysis of bacterial contamination and antibiotic residue of beef meat from city slaughterhouses in East Java Province, Indonesia. Vet. World 12 (2), 243.

Stojiljkovic, J., 2018. Bacteria of the genus Salmonella spp. In the meat and meat products. Knowl.-Int. J. 28 (4), 1285–1288.

Taib, G.A., Abdulrahman, R.F., 2022. Molecular characterization of virulence and antibiotics resistance genes and genetic diversity of salmonella enteritidis from raw chicken meat in Duhok city, Iraq. Explor. Anim. Med. Res. 12 (2).

Takaichi, M., Osawa, K., Nomoto, R., Nakanishi, N., Kameoka, M., Miura, M., Shigemura, K., Kinoshita, S., Kitagawa, K., Uda, A., Miyara, T., Mertaniasih, N.M., Hadi, U., Raharjo, D., Yulistiani, R., Fujisawa, M., Kuntaman, K., Shirakawa, T., 2022. Antibiotic resistance in non-typhoidal Salmonella enterica strains isolated from chicken meat in Indonesia. Pathogens 11 (5), 543. https://doi.org/10.3390/pathogens11050543.

Tan, C., Hazirah, M.N., Shu'aibu, I., New, C., Malcolm, T., Thung, T., Lee, E., Wendy, R., Nuzul, N., Azira, A.N., 2019. Occurrence and antibiotic resistance of Salmonella spp. in raw beef from wet market and hypermarket in Malaysia. Food Res. 3 (1), 21–27.

Tanzin, A.Z., Nath, C., Nayem, M.R.K., Sayeed, M.A., Khan, S.A., Magalhaes, R.S., Alawneh, J.I., Hassan, M.M., 2024. Detection and characterisation of colistin-resistant *Escherichia coli* in broiler meats. Microorganisms 12 (12), 2535.

Tarazi, Y.H., El-Sukhon, S.N., Ismail, Z.B., Almestarehieh, A.A., 2021. Molecular characterization of enterohemorrhagic *Escherichia coli* isolated from diarrhea samples from human, livestock, and ground beef in North Jordan. Vet. World 14 (10), 2827.

Tasmim, S.T., Hasan, M.M., Talukder, S., Mandal, A.K., Parvin, M.S., Ali, M.Y., Ehsan, M. A., Islam, M.T., 2023. Sociodemographic determinants of use and misuse of antibiotics in commercial poultry farms in Bangladesh. IJID. Reg. 7, 146–158. https://doi.org/10.1016/j.ijregi.2023.01.001.

Thapa, S.P., Shrestha, S., Anal, A.K., 2020. Addressing the antibiotic resistance and improving the food safety in food supply chain (farm-to-fork) in Southeast Asia. Food Control 108, 106809.

Theocharidi, N.A., Balta, I., Houhoula, D., Tsantes, A.G., Lalliotis, G.P., Polydera, A.C., Stamatis, H., Halvatsiotis, P., 2022. High prevalence of Klebsiella pneumoniae in Greek meat products: detection of virulence and antimicrobial resistance genes by molecular techniques. Foods (5), 11. https://doi.org/10.3390/foods11050708.

Thung, T., Mahyudin, N.A., Basri, D.F., Radzi, C.W.M., Nakaguchi, Y., Nishibuchi, M., Radu, S., 2016. Prevalence and antibiotic resistance of Salmonella enteritidis and Salmonella typhimurium in raw chicken meat at retail markets in Malaysia. Poult. Sci. 95 (8), 1888–1893.

Thung, T.Y., Radu, S., Mahyudin, N.A., Rukayadi, Y., Zakaria, Z., Mazlan, N., Tan, B.H., Lee, E., Yeoh, S.L., Chin, Y.Z., 2018. Prevalence, virulence genes and antimicrobial resistance profiles of Salmonella serovars from retail beef in Selangor, Malaysia. Front. Microbiol. 8, 2697.

Thwala, T., Madoroba, E., Basson, A., Butaye, P., 2021. Prevalence and characteristics of Staphylococcus aureus associated with meat and meat products in African countries: a review. Antibiotics 10 (9), 1108. https://www.mdpi.com/2079-6382/10/9/1108.

Trongjit, S., Angkititrakul, S., Tuttle, R.E., Poungseree, J., Padungtod, P., Chuanchuen, R., 2017. Prevalence and antimicrobial resistance in Salmonella enterica isolated from broiler chickens, pigs and meat products in Thailand–Cambodia border provinces. Microbiol. Immunol. 61 (1), 23–33.

Tyson, G.H., Nyirabahizi, E., Crarey, E., Kabera, C., Lam, C., Rice-Trujillo, C., McDermott, P.F., Tate, H., 2018. Prevalence and antimicrobial resistance of Enterococci isolated from retail meats in the United States, 2002 to 2014. Appl. Environ. Microbiol. 84 (1). https://doi.org/10.1128/aem.01902-17.

Van Boeckel, T.P., Pires, J., Silvester, R., Zhao, C., Song, J., Criscuolo, N.G., Gilbert, M., Bonhoeffer, S., Laxminarayan, R., 2019. Global trends in antimicrobial resistance in animals in low-and middle-income countries. Science (1979) 365 (6459), eaaw1944. https://doi.org/10.1126/science.aaw1944.

Van Gompel, L., Dohmen, W., Luiken, R.E.C., Bouwknegt, M., Heres, L., van Heijnsbergen, E., Jongerius-Gortemaker, B.G.M., Scherpenisse, P., Greve, G.D., Tersteeg-Zijderveld, M.H.G., Wadepohl, K., Ribeiro Duarte, A.S., Muñoz-Gómez, V., Fischer, J., Skarżyńska, M., Wasyl, D., Wagenaar, J.A., Urlings, B.A.P., Dorado-García, A., Wouters, I.M., Heederik, D.J.J., Schmitt, H., Smit, L.A.M., 2019. Occupational exposure and carriage of antimicrobial resistance genes (tetW, ermB) in pig slaughterhouse workers. Ann. Work Expo Health 64 (2), 125–137. https://doi.org/10.1093/annweh/wxz098.

Wang, X., Zhang, Y., Li, C., Li, G., Wu, D., Li, T., Qu, Y., Deng, W., He, Y., Penttinen, P., Zhang, H., Huang, Y., Zhao, K., Zou, L., 2022. Antimicrobial resistance of *Escherichia coli*, Enterobacter spp., Klebsiella pneumoniae and Enterococcus spp. Isolated from the feces of giant panda. BMC. Microbiol. 22 (1), 102. https://doi.org/10.1186/s12866-022-02514-0.

Wardhana, D.K., Haskito, A.E.P., Purnama, M.T.E., Safitri, D.A., Annisa, S., 2021. Detection of microbial contamination in chicken meat from local markets in Surabaya, East Java, Indonesia. Vet. World 14 (12), 3138.

Weerasooriya, G., Dulakshi, H.M.T., de Alwis, P.S., Bandara, S., Premarathne, K.R.P.S., Dissanayake, N., Liyanagunawardena, N., Wijemuni, M.I., Priyantha, M.A.R., 2024. Persistence of Salmonella and Campylobacter on whole chicken carcasses under the different chlorine concentrations used in the chill tank of processing plants in Sri Lanka. Pathogens 13 (8), 664. https://doi.org/10.3390/pathogens13080664.

Wei, B., Cha, S.-Y., Yoon, R.-H., Kang, M., Roh, J.-H., Seo, H.-S., Lee, J.-A., Jang, H.-K., 2016. Prevalence and antimicrobial resistance of *Campylobacter* spp. Isolated from retail chicken and duck meat in South Korea. Food Control 62, 63–68.

Wickham, H., 2011. ggplot2. Wiley Interdiscip. Rev.: Comput. Stat. 3 (2), 180–185.

Wu, B., Ed-Dra, A., Pan, H., Dong, C., Jia, C., Yue, M., 2021. Genomic investigation of Salmonella isolates recovered from a pig slaughtering process in Hangzhou, China [original research]. Front. Microbiol. 12, 2021. https://doi.org/10.3389/fmicb.2021.704636.

Wu, S., Huang, J., Wu, Q., Zhang, J., Zhang, F., Yang, X., Wu, H., Zeng, H., Chen, M., Ding, Y., Wang, J., Lei, T., Zhang, S., Xue, L., 2018. Staphylococcus aureus isolated from retail meat and meat products in China: incidence, antibiotic resistance and genetic diversity [original research]. Front. Microbiol. 9, 2018. https://doi.org/10.3389/fmicb.2018.02767.

Xavier, C., Gonzales-Barron, U., Paula, V., Estevinho, L., Cadavez, V., 2014. Meta-analysis of the incidence of foodborne pathogens in Portuguese meats and their products. Food Res. Int. 55, 311–323.

Xaymountry, S., Chukanhom, K., Klangair, S., Jiwakanon, J., Angkititrakul, S., 2022. Seasonal and antimicrobial resistance distributions of Salmonella isolated from pork, beef and chicken meat in Vientiane Capital, Lao PDR. Thai J. Vet. Med. 52 (1), 57–62.

Yam, E.L.Y., Hsu, L.Y., Yap, E.P.-H., Yeo, T.W., Lee, V., Schlundt, J., Lwin, M.O., Limmathurotsakul, D., Jit, M., Dedon, P., 2019. In: Proceedings of the Antimicrobial Resistance in the Asia Pacific region: A Meeting Report. Springer.

Yang, X., Huang, J., Zhang, Y., Liu, S., Chen, L., Xiao, C., Zeng, H., Wei, X., Gu, Q., Li, Y., Wang, J., Ding, Y., Zhang, J., Wu, Q., 2020. Prevalence, abundance, serovars and antimicrobial resistance of Salmonella isolated from retail raw poultry meat in China. Sci. Total Environ. 713, 136385. https://doi.org/10.1016/j.scitotenv.2019.136385.

Yang, X., Wu, Q., Zhang, J., Huang, J., Chen, L., Wu, S., Zeng, H., Wang, J., Chen, M., Wu, H., Gu, Q., Wei, X., 2019. Prevalence, bacterial load, and antimicrobial resistance of Salmonella serovars isolated from retail meat and meat products in China. Front. Microbiol. 10, 2121. https://doi.org/10.3389/fmicb.2019.02121.

Yang, Y., Cai, S., Mo, C., Dong, J., Chen, S., Wen, Z., 2025. Profiles of antibiotic resistome risk in diverse water environments. Commun. Earth. Environ. 6 (1), 158.

Yu, T., Jiang, X., Fu, K., Liu, B., Xu, D., Ji, S., Zhou, L., 2015. Detection of extended-spectrum β-lactamase and plasmid-mediated quinolone resistance determinants in *Escherichia coli* isolates from retail meat in China. J. Food Sci. 80 (5), M1039–M1043. https://doi.org/10.1111/1750-3841.12870.

Yu, T., Zhang, J., Jiang, X., Wu, J., Dai, Z., Wu, Z., Liang, Y., Wang, X., 2016. Characterization and horizontal transfer of class 1 integrons in *Escherichia coli* isolates from cooked meat products. J. Infect. Dev. Ctries 10 (1), 68–73. https://doi.org/10.3855/jidc.6858.

Zaiko, E.V., Bataeva, D.S., Yushina, Y.K., Grudistova, M.A., Velebit, B., 2021. Prevalence, serovar, and antimicrobial resistance of Salmonella isolated from meat and minced meat used for production smoked sausage. IOP Conf. Ser.: Earth Environ. Sci. 854 (1), 012108. https://doi.org/10.1088/1755-1315/854/1/012108.

Zakaria, Z., Hassan, L., Ahmad, N., Husin, S.A., Ali, R.M., Sharif, Z., Sohaimi, N.M., Garba, B., 2021. Discerning the antimicrobial resistance, virulence, and phylogenetic relatedness of Salmonella isolates across the Human, poultry, and food materials sources in Malaysia [Original Research]. Front. Microbiol. 12, 2021. https://doi.org/10.3389/fmicb.2021.652642.

Zelalem, A., Sisay, M., Vipham, J.L., Abegaz, K., Kebede, A., Terefe, Y., 2019. The prevalence and antimicrobial resistance profiles of bacterial isolates from meat and meat products in Ethiopia: a systematic review and meta-analysis. Int. J. Food Contam. 6 (1), 1.

Zelalem, A., Sisay, M., Vipham, J.L., Abegaz, K., Kebede, A., Terefe, Y., 2019. The prevalence and antimicrobial resistance profiles of bacterial isolates from meat and meat products in Ethiopia: a systematic review and meta-analysis. Int. J. Food Contam. 6, 1–14.

Zhang, L., Fu, Y., Xiong, Z., Ma, Y., Wei, Y., Qu, X., Zhang, H., Zhang, J., Liao, M., 2018. Highly prevalent multidrug-resistant Salmonella from chicken and pork meat at retail markets in Guangdong, China [Original Research]. Front. Microbiol. 9, 2018. https://doi.org/10.3389/fmicb.2018.02104.

Zhang, T., Luo, Q., Chen, Y., Li, T., Wen, G., Zhang, R., Luo, L., Lu, Q., Ai, D., Wang, H., Shao, H., 2016. Molecular epidemiology, virulence determinants and antimicrobial resistance of Campylobacter spreading in retail chicken meat in Central China. Gut. Pathog. 8 (1), 48. https://doi.org/10.1186/s13099-016-0132-2.

Zheng, B., Huang, C., Xu, H., Guo, L., Zhang, J., Wang, X., Jiang, X., Yu, X., Jin, L., Li, X., 2020. Occurrence and genomic characterization of ESBL-producing, MCR-1-harboring *Escherichia coli* in farming soil. Front. Microbiol. 8, 2510. https://doi.org/10.3389/fmicb.2017.02510. Occurrence and Genomic Characterization of ESBL-Producing, MCR-1-Harboring *Escherichia coli* in Farming Soil Beiwen Zheng1*†. Surveying Antimicrobial Resistance: The New Complexity of the Problem, 3, 251036.

Zhou, M., Li, X., Hou, W., Wang, H., Paoli, G.C., Shi, X., 2019. Incidence and characterization of Salmonella isolates from raw meat products sold at small markets in Hubei Province, China [Original Research]. Front. Microbiol. 10, 2019. https://doi.org/10.3389/fmicb.2019.02265.

Zhou, Y., Farzana, R., Sihalath, S., Rattanavong, S., Vongsouvath, M., Mayxay, M., Sands, K., Newton, P.N., Dance, D.A., Hassan, B., 2022. A one-health sampling strategy to explore the dissemination and relationship between colistin resistance in human, animal, and environmental sectors in Laos. Engineering 15, 45–56.

Zwe, Y.H., Tang, V.C.Y., Aung, K.T., Gutiérrez, R.A., Ng, L.C., Yuk, H.-G., 2018. Prevalence, sequence types, antibiotic resistance and, gyrA mutations of Salmonella isolated from retail fresh chicken meat in Singapore. Food Control 90, 233–240. https://doi.org/10.1016/j.foodcont.2018.03.004.